

FastAPI

Reference

APIRouter class

Here's the reference information for the `APIRouter` class, with all its parameters, attributes and methods.

You can import the `APIRouter` class directly from `fastapi`:

```
from fastapi import APIRouter
```

`class` fastapi.APIRouter

```
APIRouter(  
    *,  
    prefix="",  
    tags=None,  
    dependencies=None,  
    default_response_class=Default(JSONResponse),  
    responses=None,  
    callbacks=None,  
    routes=None,  
    redirect_slashes=True,  
    default=None,  
    dependency_overrides_provider=None,  
    route_class=APIRoute,  
    on_startup=None,  
    on_shutdown=None,  
    lifespan=None,  
    deprecated=None,  
    include_in_schema=True,  
    generate_unique_id_function=Default(generate_unique_id),  
    strict_content_type=Default(True)  
)
```

Bases: `Router`

`APIRouter` class, used to group *path operations*, for example to structure an app in multiple files. It would then be included in the `FastAPI` app, or in another `APIRouter` (ultimately included in the app).

Read more about it in the [FastAPI docs for Bigger Applications - Multiple Files](#).

Example

```
from fastapi import APIRouter, FastAPI  
  
app = FastAPI()  
router = APIRouter()  
  
@router.get("/users/", tags=["users"])  
async def read_users():  
    return [{"username": "Rick"}, {"username": "Morty"}]  
  
app.include_router(router)
```

[Ask AI](#)

PARAMETER	DESCRIPTION
<code>prefix</code>	An optional path prefix for the router. TYPE: <code>str</code> DEFAULT: <code>''</code>
<code>tags</code>	A list of tags to be applied to all the <i>path operations</i> in this router. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum] None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to all the <i>path operations</i> in this router. Read more about it in the FastAPI docs for Bigger Applications - Multiple Files. TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>
<code>default_response_class</code>	The default response class to be used. Read more in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>responses</code>	Additional responses to be shown in OpenAPI. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Additional Responses in OpenAPI. And in the FastAPI docs for Bigger Applications. TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	OpenAPI callbacks that should apply to all <i>path operations</i> in this router. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>routes</code>	Note: you probably shouldn't use this parameter, it is inherited from Starlette and supported for compatibility. A list of routes to serve incoming HTTP and WebSocket requests. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>redirect_slashes</code>	Whether to detect and redirect slashes in URLs when the client doesn't use the same format. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>default</code>	Default function handler for this router. Used to handle <code>404 Not Found</code> errors. TYPE: <code>ASGIApp</code> <code>None</code> DEFAULT: <code>None</code>
<code>dependency_overrides_provider</code>	Only used internally by FastAPI to handle dependency overrides. You shouldn't need to use it. It normally points to the <code>FastAPI</code> app object. TYPE: <code>Any</code> <code>None</code> DEFAULT: <code>None</code>
<code>route_class</code>	Custom route (<i>path operation</i>) class to be used by this router. Read more about it in the FastAPI docs for Custom Request and APIRoute class. TYPE: <code>type[APIRoute]</code> DEFAULT: <code>APIRoute</code>
<code>on_startup</code>	A list of startup event handler functions. You should instead use the <code>lifespan</code> handlers. Read more in the FastAPI docs for lifespan. TYPE: <code>Sequence[Callable[[], Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>on_shutdown</code>	A list of shutdown event handler functions. You should instead use the <code>lifespan</code> handlers. Read more in the FastAPI docs for lifespan. TYPE: <code>Sequence[Callable[[], Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>lifespan</code>	A <code>Lifespan</code> context manager handler. This replaces <code>startup</code> and <code>shutdown</code> functions with a single context manager. Read more in the FastAPI docs for lifespan. TYPE: <code>Lifespan[Any]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark all <i>path operations</i> in this router as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>include_in_schema</code>	To include (or not) all the <i>path operations</i> in this router in the generated OpenAPI. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>
<code>strict_content_type</code>	Enable strict checking for request Content-Type headers. When <code>True</code> (the default), requests with a body that do not include a <code>Content-Type</code> header will not be parsed as JSON. This prevents potential cross-site request forgery (CSRF) attacks that exploit the browser's ability to send requests without a Content-Type header, bypassing CORS preflight checks. In particular applicable for apps that need to be run locally (in localhost). When <code>False</code>, requests without a <code>Content-Type</code> header will have their body parsed as JSON, which maintains compatibility with certain clients that don't send <code>Content-Type</code> headers. Read more about it in the FastAPI docs for Strict Content-Type. TYPE: <code>bool</code> DEFAULT: <code>Default(True)</code>

<> Source code in fastapi/routing.py

```

1032 def __init__(
1033     self,
1034     *,
1035     prefix: Annotated[str, Doc("An optional path prefix for the router.")] = "",
1036     tags: Annotated[
1037         list[str | Enum] | None,
1038         Doc(
1039             """
1040             A list of tags to be applied to all the *path operations* in this
1041             router.
1042
1043             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1044
1045             Read more about it in the
1046             [FastAPI docs for Path Operation Configuration]
1047             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1048             """
1049         ),
1050     ] = None,
1051     dependencies: Annotated[
1052         Sequence[params.Depends] | None,
1053         Doc(
1054             """
1055             A list of dependencies (using `Depends()`) to be applied to all the
1056             *path operations* in this router.
1057
1058             Read more about it in the
1059             [FastAPI docs for Bigger Applications - Multiple Files]
1060             (https://fastapi.tiangolo.com/tutorial/bigger-applications/#include-an-apirouter-with-a-custom-
1061             prefix-tags-responses-and-dependencies).
1062             """
1063         ),
1064     ] = None,
1065     default_response_class: Annotated[
1066         type[Response],
1067         Doc(
1068             """
1069             The default response class to be used.
1070
1071             Read more in the
1072             [FastAPI docs for Custom Response - HTML, Stream, File, others]
1073             (https://fastapi.tiangolo.com/advanced/custom-response/#default-response-class).
1074             """
1075         ),
1076     ] = Default(JSONResponse),
1077     responses: Annotated[
1078         dict[int | str, dict[str, Any]] | None,
1079         Doc(
1080             """
1081             Additional responses to be shown in OpenAPI.
1082
1083             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1084
1085             Read more about it in the
1086             [FastAPI docs for Additional Responses in OpenAPI]
1087             (https://fastapi.tiangolo.com/advanced/additional-responses/).
1088
1089             And in the
1090             [FastAPI docs for Bigger Applications](https://fastapi.tiangolo.com/tutorial/bigger-
1091             applications/#include-an-apirouter-with-a-custom-prefix-tags-responses-and-dependencies).
1092             """
1093         ),
1094     ] = None,
1095     callbacks: Annotated[
1096         list[BaseRoute] | None,
1097         Doc(
1098             """
1099             OpenAPI callbacks that should apply to all *path operations* in this
1100             router.
1101
1102             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1103             """

```

```

1104         Read more about it in the
1105         [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
1106         callbacks/).
1107         """
1108     ),
1109     ] = None,
1110     routes: Annotated[
1111         list[BaseRoute] | None,
1112         Doc(
1113             """
1114             **Note**: you probably shouldn't use this parameter, it is inherited
1115             from Starlette and supported for compatibility.
1116
1117             ---
1118
1119             A list of routes to serve incoming HTTP and WebSocket requests.
1120             """
1121         ),
1122         deprecated(
1123             """
1124             You normally wouldn't use this parameter with FastAPI, it is inherited
1125             from Starlette and supported for compatibility.
1126
1127             In FastAPI, you normally would use the *path operation methods*,
1128             like `router.get()`, `router.post()`, etc.
1129             """
1130         ),
1131     ] = None,
1132     redirect_slashes: Annotated[
1133         bool,
1134         Doc(
1135             """
1136             Whether to detect and redirect slashes in URLs when the client doesn't
1137             use the same format.
1138             """
1139         ),
1140     ] = True,
1141     default: Annotated[
1142         ASGIApp | None,
1143         Doc(
1144             """
1145             Default function handler for this router. Used to handle
1146             404 Not Found errors.
1147             """
1148         ),
1149     ] = None,
1150     dependency_overrides_provider: Annotated[
1151         Any | None,
1152         Doc(
1153             """
1154             Only used internally by FastAPI to handle dependency overrides.
1155
1156             You shouldn't need to use it. It normally points to the `FastAPI` app
1157             object.
1158             """
1159         ),
1160     ] = None,
1161     route_class: Annotated[
1162         type[APIRoute],
1163         Doc(
1164             """
1165             Custom route (*path operation*) class to be used by this router.
1166
1167             Read more about it in the
1168             [FastAPI docs for Custom Request and APIRoute class]
1169             (https://fastapi.tiangolo.com/how-to/custom-request-and-route/#custom-apiroute-class-in-a-
1170             router).
1171             """
1172         ),
1173     ] = APIRoute,
1174     on_startup: Annotated[
1175         Sequence[Callable[[], Any]] | None,
1176         Doc(
1177             """
1178             A list of startup event handler functions.
1179

```

```

1180         You should instead use the `lifespan` handlers.
1181
1182         Read more in the [FastAPI docs for `lifespan`]
1183         (https://fastapi.tiangolo.com/advanced/events/).
1184         """
1185     ),
1186     ] = None,
1187     on_shutdown: Annotated[
1188         Sequence[Callable[[], Any]] | None,
1189         Doc(
1190             """
1191             A list of shutdown event handler functions.
1192
1193             You should instead use the `lifespan` handlers.
1194
1195             Read more in the
1196             [FastAPI docs for `lifespan`](https://fastapi.tiangolo.com/advanced/events/).
1197             """
1198         ),
1199     ] = None,
1200     # the generic to Lifespan[AppType] is the type of the top level application
1201     # which the router cannot know statically, so we use typing.Any
1202     lifespan: Annotated[
1203         Lifespan[Any] | None,
1204         Doc(
1205             """
1206             A `Lifespan` context manager handler. This replaces `startup` and
1207             `shutdown` functions with a single context manager.
1208
1209             Read more in the
1210             [FastAPI docs for `lifespan`](https://fastapi.tiangolo.com/advanced/events/).
1211             """
1212         ),
1213     ] = None,
1214     deprecated: Annotated[
1215         bool | None,
1216         Doc(
1217             """
1218             Mark all *path operations* in this router as deprecated.
1219
1220             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1221
1222             Read more about it in the
1223             [FastAPI docs for Path Operation Configuration]
1224             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1225             """
1226         ),
1227     ] = None,
1228     include_in_schema: Annotated[
1229         bool,
1230         Doc(
1231             """
1232             To include (or not) all the *path operations* in this router in the
1233             generated OpenAPI.
1234
1235             This affects the generated OpenAPI (e.g. visible at `/docs`).
1236
1237             Read more about it in the
1238             [FastAPI docs for Query Parameters and String Validations]
1239             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-openapi).
1240             """
1241         ),
1242     ] = True,
1243     generate_unique_id_function: Annotated[
1244         Callable[[APIRoute], str],
1245         Doc(
1246             """
1247             Customize the function used to generate unique IDs for the *path
1248             operations* shown in the generated OpenAPI.
1249
1250             This is particularly useful when automatically generating clients or
1251             SDKs for your API.
1252
1253             Read more about it in the
1254
1255             """

```

```

1256         [FastAPI docs about how to Generate Clients]
1257         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
1258         """
1259     ),
1260     ] = Default(generate_unique_id),
1261     strict_content_type: Annotated[
1262         bool,
1263         Doc(
1264             """
1265             Enable strict checking for request Content-Type headers.
1266
1267             When `True` (the default), requests with a body that do not include
1268             a `Content-Type` header will not be parsed as JSON.
1269
1270             This prevents potential cross-site request forgery (CSRF) attacks
1271             that exploit the browser's ability to send requests without a
1272             Content-Type header, bypassing CORS preflight checks. In particular
1273             applicable for apps that need to be run locally (in localhost).
1274
1275             When `False`, requests without a `Content-Type` header will have
1276             their body parsed as JSON, which maintains compatibility with
1277             certain clients that don't send `Content-Type` headers.
1278
1279             Read more about it in the
1280             [FastAPI docs for Strict Content-Type](https://fastapi.tiangolo.com/advanced/strict-
1281             content-type/).
1282             """
1283         ),
1284     ] = Default(True),
1285 ) -> None:
1286     # Determine the lifespan context to use
1287     if lifespan is None:
1288         # Use the default lifespan that runs on_startup/on_shutdown handlers
1289         lifespan_context: Lifespan[Any] = _DefaultLifespan(self)
1290     elif inspect.isasyncgenfunction(lifespan):
1291         lifespan_context = asynccontextmanager(lifespan)
1292     elif inspect.isgeneratorfunction(lifespan):
1293         lifespan_context = _wrap_gen_lifespan_context(lifespan)
1294     else:
1295         lifespan_context = lifespan
1296     self.lifespan_context = lifespan_context
1297
1298     super().__init__(
1299         routes=routes,
1300         redirect_slashes=redirect_slashes,
1301         default=default,
1302         lifespan=lifespan_context,
1303     )
1304     if prefix:
1305         assert prefix.startswith("/"), "A path prefix must start with '/'"
1306         assert not prefix.endswith("/"), (
1307             "A path prefix must not end with '/', as the routes will start with '/'"
1308         )
1309
1310     # Handle on_startup/on_shutdown locally since Starlette removed support
1311     # Ref: https://github.com/Kludex/starlette/pull/3117
1312     # TODO: deprecate this once the lifespan (or alternative) interface is improved
1313     self.on_startup: list[Callable[[], Any]] = (
1314         [] if on_startup is None else list(on_startup)
1315     )
1316     self.on_shutdown: list[Callable[[], Any]] = (
1317         [] if on_shutdown is None else list(on_shutdown)
1318     )
1319
1320     self.prefix = prefix
1321     self.tags: list[str | Enum] = tags or []
1322     self.dependencies = list(dependencies or [])
1323     self.deprecated = deprecated
1324     self.include_in_schema = include_in_schema
1325     self.responses = responses or {}
1326     self.callbacks = callbacks or []
1327     self.dependency_overrides_provider = dependency_overrides_provider
1328     self.route_class = route_class
1329     self.default_response_class = default_response_class
1330     self.generate_unique_id_function = generate_unique_id_function
1331     self.strict_content_type = strict_content_type

```


meth websocket

```
websocket(path, name=None, *, dependencies=None)
```

Decorate a WebSocket function.

Read more about it in the [FastAPI docs for WebSockets](#).

Example

EXAMPLE

```
from fastapi import APIRouter, FastAPI, WebSocket

app = FastAPI()
router = APIRouter()

@router.websocket("/ws")
async def websocket_endpoint(websocket: WebSocket):
    await websocket.accept()
    while True:
        data = await websocket.receive_text()
        await websocket.send_text(f"Message text was: {data}")

app.include_router(router)
```

PARAMETER	DESCRIPTION
path	WebSocket path. TYPE: str
name	A name for the WebSocket. Only used internally. TYPE: str None DEFAULT: None
dependencies	A list of dependencies (using Depends()) to be used for this WebSocket. Read more about it in the FastAPI docs for WebSockets . TYPE: Sequence[Depends] None DEFAULT: None

<> Source code in `fastapi/routing.py`

```

1502 def websocket(
1503     self,
1504     path: Annotated[
1505         str,
1506         Doc(
1507             """
1508             WebSocket path.
1509             """
1510         ),
1511     ],
1512     name: Annotated[
1513         str | None,
1514         Doc(
1515             """
1516             A name for the WebSocket. Only used internally.
1517             """
1518         ),
1519     ] = None,
1520     *,
1521     dependencies: Annotated[
1522         Sequence[params.Depends] | None,
1523         Doc(
1524             """
1525             A list of dependencies (using `Depends()`) to be used for this
1526             WebSocket.
1527
1528             Read more about it in the
1529             [FastAPI docs for WebSockets](https://fastapi.tiangolo.com/advanced/websockets/).
1530             """
1531         ),
1532     ] = None,
1533 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
1534     """
1535     Decorate a WebSocket function.
1536
1537     Read more about it in the
1538     [FastAPI docs for WebSockets](https://fastapi.tiangolo.com/advanced/websockets/).
1539
1540     **Example**
1541
1542     ## Example
1543
1544     ```python
1545     from fastapi import APIRouter, FastAPI, WebSocket
1546
1547     app = FastAPI()
1548     router = APIRouter()
1549
1550     @router.websocket("/ws")
1551     async def websocket_endpoint(websocket: WebSocket):
1552         await websocket.accept()
1553         while True:
1554             data = await websocket.receive_text()
1555             await websocket.send_text(f"Message text was: {data}")
1556
1557     app.include_router(router)
1558     ```
1559     """
1560
1561     def decorator(func: DecoratedCallable) -> DecoratedCallable:
1562         self.add_api_websocket_route(
1563             path, func, name=name, dependencies=dependencies
1564         )
1565         return func
1566
1567     return decorator

```

`meth` include_router

```
include_router(
    router,
    *,
    prefix="",
    tags=None,
    dependencies=None,
    default_response_class=Default(JSONResponse),
    responses=None,
    callbacks=None,
    deprecated=None,
    include_in_schema=True,
    generate_unique_id_function=Default(generate_unique_id)
)
```

Include another `APIRouter` in the same current `APIRouter`.

Read more about it in the [FastAPI docs for Bigger Applications](#).

EXAMPLE

```
from fastapi import APIRouter, FastAPI

app = FastAPI()
internal_router = APIRouter()
users_router = APIRouter()

@users_router.get("/users/")
def read_users():
    return [{"name": "Rick"}, {"name": "Morty"}]

internal_router.include_router(users_router)
app.include_router(internal_router)
```

PARAMETER	DESCRIPTION
<code>router</code>	The <code>APIRouter</code> to include. TYPE: <code>APIRouter</code>
<code>prefix</code>	An optional path prefix for the router. TYPE: <code>str</code> DEFAULT: <code>''</code>
<code>tags</code>	A list of tags to be applied to all the <i>path operations</i> in this router. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration . TYPE: <code>list[str Enum] None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to all the <i>path operations</i> in this router. Read more about it in the FastAPI docs for Bigger Applications - Multiple Files . TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>default_response_class</code>	The default response class to be used. Read more in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>responses</code>	Additional responses to be shown in OpenAPI. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Additional Responses in OpenAPI. And in the FastAPI docs for Bigger Applications. TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	OpenAPI callbacks that should apply to all <i>path operations</i> in this router. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark all <i>path operations</i> in this router as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>bool None</code> DEFAULT: <code>None</code>
<code>include_in_schema</code>	Include (or not) all the <i>path operations</i> in this router in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in `fastapi/routing.py`

```

1578 def include_router(
1579     self,
1580     router: Annotated["APIRouter", Doc("The `APIRouter` to include.")],
1581     *,
1582     prefix: Annotated[str, Doc("An optional path prefix for the router.")],
1583     tags: Annotated[
1584         list[str | Enum] | None,
1585         Doc(
1586             """
1587             A list of tags to be applied to all the *path operations* in this
1588             router.
1589
1590             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1591
1592             Read more about it in the
1593             [FastAPI docs for Path Operation Configuration]
1594             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1595             """
1596         ),
1597     ] = None,
1598     dependencies: Annotated[
1599         Sequence[params.Depends] | None,
1600         Doc(
1601             """
1602             A list of dependencies (using `Depends()`) to be applied to all the
1603             *path operations* in this router.
1604
1605             Read more about it in the
1606             [FastAPI docs for Bigger Applications - Multiple Files]
1607             (https://fastapi.tiangolo.com/tutorial/bigger-applications/#include-an-apirouter-with-a-custom-
1608             prefix-tags-responses-and-dependencies).
1609             """
1610         ),
1611     ] = None,
1612     default_response_class: Annotated[
1613         type[Response],
1614         Doc(
1615             """
1616             The default response class to be used.
1617
1618             Read more in the
1619             [FastAPI docs for Custom Response - HTML, Stream, File, others]
1620             (https://fastapi.tiangolo.com/advanced/custom-response/#default-response-class).
1621             """
1622         ),
1623     ] = Default(JSONResponse),
1624     responses: Annotated[
1625         dict[int | str, dict[str, Any]] | None,
1626         Doc(
1627             """
1628             Additional responses to be shown in OpenAPI.
1629
1630             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1631
1632             Read more about it in the
1633             [FastAPI docs for Additional Responses in OpenAPI]
1634             (https://fastapi.tiangolo.com/advanced/additional-responses/).
1635
1636             And in the
1637             [FastAPI docs for Bigger Applications](https://fastapi.tiangolo.com/tutorial/bigger-
1638             applications/#include-an-apirouter-with-a-custom-prefix-tags-responses-and-dependencies).
1639             """
1640         ),
1641     ] = None,
1642     callbacks: Annotated[
1643         list[BaseRoute] | None,
1644         Doc(
1645             """
1646             OpenAPI callbacks that should apply to all *path operations* in this
1647             router.
1648
1649             It will be added to the generated OpenAPI (e.g. visible at `/docs`).

```

```

1650
1651         Read more about it in the
1652         [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
1653         callbacks/).
1654         """
1655     ),
1656     ] = None,
1657     deprecated: Annotated[
1658         bool | None,
1659         Doc(
1660             """
1661             Mark all *path operations* in this router as deprecated.
1662
1663             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1664
1665             Read more about it in the
1666             [FastAPI docs for Path Operation Configuration]
1667             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1668             """
1669         ),
1670     ] = None,
1671     include_in_schema: Annotated[
1672         bool,
1673         Doc(
1674             """
1675             Include (or not) all the *path operations* in this router in the
1676             generated OpenAPI schema.
1677
1678             This affects the generated OpenAPI (e.g. visible at `/docs`).
1679             """
1680         ),
1681     ] = True,
1682     generate_unique_id_function: Annotated[
1683         Callable[[APIRoute], str],
1684         Doc(
1685             """
1686             Customize the function used to generate unique IDs for the *path
1687             operations* shown in the generated OpenAPI.
1688
1689             This is particularly useful when automatically generating clients or
1690             SDKs for your API.
1691
1692             Read more about it in the
1693             [FastAPI docs about how to Generate Clients]
1694             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
1695             """
1696         ),
1697     ] = Default(generate_unique_id),
1698 ) -> None:
1699     """
1700     Include another `APIRouter` in the same current `APIRouter`.
1701
1702     Read more about it in the
1703     [FastAPI docs for Bigger Applications](https://fastapi.tiangolo.com/tutorial/bigger-
1704     applications/).
1705
1706     ## Example
1707
1708     ```python
1709     from fastapi import APIRouter, FastAPI
1710
1711     app = FastAPI()
1712     internal_router = APIRouter()
1713     users_router = APIRouter()
1714
1715     @users_router.get("/users/")
1716     def read_users():
1717         return [{"name": "Rick"}, {"name": "Morty"}]
1718
1719     internal_router.include_router(users_router)
1720     app.include_router(internal_router)
1721     """
1722
1723     assert self is not router, (
1724         "Cannot include the same APIRouter instance into itself. "
1725         "Did you mean to include a different router?"

```

```

1726     )
1727     if prefix:
1728         assert prefix.startswith("/"), "A path prefix must start with '/'"
1729         assert not prefix.endswith("/"), (
1730             "A path prefix must not end with '/', as the routes will start with '/'"
1731         )
1732     else:
1733         for r in router.routes:
1734             path = getattr(r, "path") # noqa: B009
1735             name = getattr(r, "name", "unknown")
1736             if path is not None and not path:
1737                 raise FastAPIError(
1738                     f"Prefix and path cannot be both empty (path operation: {name})"
1739                 )
1740     if responses is None:
1741         responses = {}
1742     for route in router.routes:
1743         if isinstance(route, APIRoute):
1744             combined_responses = {**responses, **route.responses}
1745             use_response_class = get_value_or_default(
1746                 route.response_class,
1747                 router.default_response_class,
1748                 default_response_class,
1749                 self.default_response_class,
1750             )
1751             current_tags = []
1752             if tags:
1753                 current_tags.extend(tags)
1754             if route.tags:
1755                 current_tags.extend(route.tags)
1756             current_dependencies: list[params.Depends] = []
1757             if dependencies:
1758                 current_dependencies.extend(dependencies)
1759             if route.dependencies:
1760                 current_dependencies.extend(route.dependencies)
1761             current_callbacks = []
1762             if callbacks:
1763                 current_callbacks.extend(callbacks)
1764             if route.callbacks:
1765                 current_callbacks.extend(route.callbacks)
1766             current_generate_unique_id = get_value_or_default(
1767                 route.generate_unique_id_function,
1768                 router.generate_unique_id_function,
1769                 generate_unique_id_function,
1770                 self.generate_unique_id_function,
1771             )
1772             self.add_api_route(
1773                 prefix + route.path,
1774                 route.endpoint,
1775                 response_model=route.response_model,
1776                 status_code=route.status_code,
1777                 tags=current_tags,
1778                 dependencies=current_dependencies,
1779                 summary=route.summary,
1780                 description=route.description,
1781                 response_description=route.response_description,
1782                 responses=combined_responses,
1783                 deprecated=route.deprecated or deprecated or self.deprecated,
1784                 methods=route.methods,
1785                 operation_id=route.operation_id,
1786                 response_model_include=route.response_model_include,
1787                 response_model_exclude=route.response_model_exclude,
1788                 response_model_by_alias=route.response_model_by_alias,
1789                 response_model_exclude_unset=route.response_model_exclude_unset,
1790                 response_model_exclude_defaults=route.response_model_exclude_defaults,
1791                 response_model_exclude_none=route.response_model_exclude_none,
1792                 include_in_schema=route.include_in_schema
1793                 and self.include_in_schema
1794                 and include_in_schema,
1795                 response_class=use_response_class,
1796                 name=route.name,
1797                 route_class_override=type(route),
1798                 callbacks=current_callbacks,
1799                 openapi_extra=route.openapi_extra,
1800                 generate_unique_id_function=current_generate_unique_id,
1801                 strict_content_type=get_value_or_default(

```

```

1802         route.strict_content_type,
1803         router.strict_content_type,
1804         self.strict_content_type,
1805     ),
1806 )
1807 elif isinstance(route, routing.Route):
1808     methods = list(route.methods or [])
1809     self.add_route(
1810         prefix + route.path,
1811         route.endpoint,
1812         methods=methods,
1813         include_in_schema=route.include_in_schema,
1814         name=route.name,
1815     )
1816 elif isinstance(route, APIWebSocketRoute):
1817     current_dependencies = []
1818     if dependencies:
1819         current_dependencies.extend(dependencies)
1820     if route.dependencies:
1821         current_dependencies.extend(route.dependencies)
1822     self.add_api_websocket_route(
1823         prefix + route.path,
1824         route.endpoint,
1825         dependencies=current_dependencies,
1826         name=route.name,
1827     )
1828 elif isinstance(route, routing.WebSocketRoute):
1829     self.add_websocket_route(
1830         prefix + route.path, route.endpoint, name=route.name
1831     )
1832 for handler in router.on_startup:
1833     self.add_event_handler("startup", handler)
1834 for handler in router.on_shutdown:
1835     self.add_event_handler("shutdown", handler)
1836 self.lifespan_context = _merge_lifespan_context(
1837     self.lifespan_context,
1838     router.lifespan_context,
1839 )

```

meth get

```

get(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP GET operation.

EXAMPLE

```
from fastapi import APIRouter, FastAPI

app = FastAPI()
router = APIRouter()

@router.get("/items/")
def read_items():
    return [{"name": "Empanada"}, {"name": "Arepa"}]

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: <code>Any</code> DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum] None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API.

PARAMETER	DESCRIPTION
	You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense.

PARAMETER	DESCRIPTION
	Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

1831 def get(
1832     self,
1833     path: Annotated[
1834         str,
1835         Doc(
1836             """
1837             The URL path to be used for this *path operation*.
1838
1839             For example, in `http://example.com/items`, the path is `/items`.
1840             """
1841         ),
1842     ],
1843     *,
1844     response_model: Annotated[
1845         Any,
1846         Doc(
1847             """
1848             The type to use for the response.
1849
1850             It could be any valid Pydantic *field* type. So, it doesn't have to
1851             be a Pydantic model, it could be other things, like a `list`, `dict`,
1852             etc.
1853
1854             It will be used for:
1855
1856             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
1857               show it as the response (JSON Schema).
1858             * Serialization: you could return an arbitrary object and the
1859               `response_model` would be used to serialize that object into the
1860               corresponding JSON.
1861             * Filtering: the JSON sent to the client will only contain the data
1862               (fields) defined in the `response_model`. If you returned an object
1863               that contains an attribute `password` but the `response_model` does
1864               not include that field, the JSON sent to the client would not have
1865               that `password`.
1866             * Validation: whatever you return will be serialized with the
1867               `response_model`, converting any data as necessary to generate the
1868               corresponding JSON. But if the data in the object returned is not
1869               valid, that would mean a violation of the contract with the client,
1870               so it's an error from the API developer. So, FastAPI will raise an
1871               error and return a 500 error code (Internal Server Error).
1872
1873               Read more about it in the
1874               [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
1875               model/).
1876             """
1877         ),
1878     ] = Default(None),
1879     status_code: Annotated[
1880         int | None,
1881         Doc(
1882             """
1883             The default status code to be used for the response.
1884
1885             You could override the status code by returning a response directly.
1886
1887             Read more about it in the
1888             [FastAPI docs for Response Status Code]
1889             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
1890             """
1891         ),
1892     ] = None,
1893     tags: Annotated[
1894         list[str | Enum] | None,
1895         Doc(
1896             """
1897             A list of tags to be applied to the *path operation*.
1898
1899             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1900
1901             Read more about it in the
1902 
```

```

1903         [FastAPI docs for Path Operation Configuration]
1904         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
1905         """
1906     ),
1907     ] = None,
1908     dependencies: Annotated[
1909         Sequence[params.Depends] | None,
1910         Doc(
1911             """
1912             A list of dependencies (using `Depends()`) to be applied to the
1913             *path operation*.
1914
1915             Read more about it in the
1916             [FastAPI docs for Dependencies in path operation decorators]
1917             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
1918             """
1919         ),
1920     ] = None,
1921     summary: Annotated[
1922         str | None,
1923         Doc(
1924             """
1925             A summary for the *path operation*.
1926
1927             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1928
1929             Read more about it in the
1930             [FastAPI docs for Path Operation Configuration]
1931             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1932             """
1933         ),
1934     ] = None,
1935     description: Annotated[
1936         str | None,
1937         Doc(
1938             """
1939             A description for the *path operation*.
1940
1941             If not provided, it will be extracted automatically from the docstring
1942             of the *path operation function*.
1943
1944             It can contain Markdown.
1945
1946             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1947
1948             Read more about it in the
1949             [FastAPI docs for Path Operation Configuration]
1950             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1951             """
1952         ),
1953     ] = None,
1954     response_description: Annotated[
1955         str,
1956         Doc(
1957             """
1958             The description for the default response.
1959
1960             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1961             """
1962         ),
1963     ] = "Successful Response",
1964     responses: Annotated[
1965         dict[int | str, dict[str, Any]] | None,
1966         Doc(
1967             """
1968             Additional responses that could be returned by this *path operation*.
1969
1970             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1971             """
1972         ),
1973     ] = None,
1974     deprecated: Annotated[
1975         bool | None,
1976         Doc(
1977             """
1978             Mark this *path operation* as deprecated.

```

```

1979         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1980         """
1981     ),
1982 ] = None,
1983 operation_id: Annotated[
1984     str | None,
1985     Doc(
1986         """
1987         Custom operation ID to be used by this *path operation*.
1988
1989         By default, it is generated automatically.
1990
1991         If you provide a custom operation ID, you need to make sure it is
1992         unique for the whole API.
1993
1994         You can customize the
1995         operation ID generation with the parameter
1996         `generate_unique_id_function` in the `FastAPI` class.
1997
1998         Read more about it in the
1999         [FastAPI docs about how to Generate Clients]
2000         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2001         """
2002     ),
2003 ] = None,
2004 response_model_include: Annotated[
2005     IncEx | None,
2006     Doc(
2007         """
2008         Configuration passed to Pydantic to include only certain fields in the
2009         response data.
2010
2011         Read more about it in the
2012         [FastAPI docs for Response Model - Return Type]
2013         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2014         """
2015     ),
2016 ] = None,
2017 response_model_exclude: Annotated[
2018     IncEx | None,
2019     Doc(
2020         """
2021         Configuration passed to Pydantic to exclude certain fields in the
2022         response data.
2023
2024         Read more about it in the
2025         [FastAPI docs for Response Model - Return Type]
2026         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2027         """
2028     ),
2029 ] = None,
2030 response_model_by_alias: Annotated[
2031     bool,
2032     Doc(
2033         """
2034         Configuration passed to Pydantic to define if the response model
2035         should be serialized by alias when an alias is used.
2036
2037         Read more about it in the
2038         [FastAPI docs for Response Model - Return Type]
2039         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2040         """
2041     ),
2042 ] = True,
2043 response_model_exclude_unset: Annotated[
2044     bool,
2045     Doc(
2046         """
2047         Configuration passed to Pydantic to define if the response data
2048         should have all the fields, including the ones that were not set and
2049         have their default values. This is different from
2050         `response_model_exclude_defaults` in that if the fields are set,

```

```

2055         they will be included in the response, even if the value is the same
2056         as the default.
2057
2058         When `True`, default values are omitted from the response.
2059
2060         Read more about it in the
2061         [FastAPI docs for Response Model - Return Type]
2062         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2063         parameter).
2064         """
2065     ),
2066     ] = False,
2067     response_model_exclude_defaults: Annotated[
2068         bool,
2069         Doc(
2070             """
2071             Configuration passed to Pydantic to define if the response data
2072             should have all the fields, including the ones that have the same value
2073             as the default. This is different from `response_model_exclude_unset`
2074             in that if the fields are set but contain the same default values,
2075             they will be excluded from the response.
2076
2077             When `True`, default values are omitted from the response.
2078
2079             Read more about it in the
2080             [FastAPI docs for Response Model - Return Type]
2081             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2082             parameter).
2083             """
2084         ),
2085     ] = False,
2086     response_model_exclude_none: Annotated[
2087         bool,
2088         Doc(
2089             """
2090             Configuration passed to Pydantic to define if the response data should
2091             exclude fields set to `None`.
2092
2093             This is much simpler (less smart) than `response_model_exclude_unset`
2094             and `response_model_exclude_defaults`. You probably want to use one of
2095             those two instead of this one, as those allow returning `None` values
2096             when it makes sense.
2097
2098             Read more about it in the
2099             [FastAPI docs for Response Model - Return Type]
2100             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
2101             """
2102         ),
2103     ] = False,
2104     include_in_schema: Annotated[
2105         bool,
2106         Doc(
2107             """
2108             Include this *path operation* in the generated OpenAPI schema.
2109
2110             This affects the generated OpenAPI (e.g. visible at `/docs`).
2111
2112             Read more about it in the
2113             [FastAPI docs for Query Parameters and String Validations]
2114             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
2115             openapi).
2116             """
2117         ),
2118     ] = True,
2119     response_class: Annotated[
2120         type[Response],
2121         Doc(
2122             """
2123             Response class to be used for this *path operation*.
2124
2125             This will not be used if you return a response directly.
2126
2127             Read more about it in the
2128             [FastAPI docs for Custom Response - HTML, Stream, File, others]
2129             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
2130             """

```



```

2131     ),
2132     ] = Default(JSONResponse),
2133     name: Annotated[
2134         str | None,
2135         Doc(
2136             """
2137             Name for this *path operation*. Only used internally.
2138             """
2139         ),
2140     ] = None,
2141     callbacks: Annotated[
2142         list[BaseRoute] | None,
2143         Doc(
2144             """
2145             List of *path operations* that will be used as OpenAPI callbacks.
2146
2147             This is only for OpenAPI documentation, the callbacks won't be used
2148             directly.
2149
2150             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2151
2152             Read more about it in the
2153             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
2154             callbacks/).
2155             """
2156         ),
2157     ] = None,
2158     openapi_extra: Annotated[
2159         dict[str, Any] | None,
2160         Doc(
2161             """
2162             Extra metadata to be included in the OpenAPI schema for this *path
2163             operation*.
2164
2165             Read more about it in the
2166             [FastAPI docs for Path Operation Advanced Configuration]
2167             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
2168             path-operation-schema).
2169             """
2170         ),
2171     ] = None,
2172     generate_unique_id_function: Annotated[
2173         Callable[[APIRoute], str],
2174         Doc(
2175             """
2176             Customize the function used to generate unique IDs for the *path
2177             operations* shown in the generated OpenAPI.
2178
2179             This is particularly useful when automatically generating clients or
2180             SDKs for your API.
2181
2182             Read more about it in the
2183             [FastAPI docs about how to Generate Clients]
2184             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2185             """
2186         ),
2187     ] = Default(generate_unique_id),
2188 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
2189     """
2190     Add a *path operation* using an HTTP GET operation.
2191
2192     ## Example
2193
2194     ```python
2195     from fastapi import APIRouter, FastAPI
2196
2197     app = FastAPI()
2198     router = APIRouter()
2199
2200     @router.get("/items/")
2201     def read_items():
2202         return [{"name": "Empanada"}, {"name": "Arepa"}]
2203
2204     app.include_router(router)
2205     """
2206

```

```

    return self.api_route(
        path=path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        methods=["GET"],
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth put

```

put(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP PUT operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI
from pydantic import BaseModel

class Item(BaseModel):
    name: str
    description: str | None = None

app = FastAPI()
router = APIRouter()

```

```
@router.put("/items/{item_id}")
def replace_item(item_id: str, item: Item):
    return {"message": "Item replaced", "id": item_id}

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> <code>None</code> DEFAULT: <code>None</code>
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

2208 def put(
2209     self,
2210     path: Annotated[
2211         str,
2212         Doc(
2213             """
2214             The URL path to be used for this *path operation*.
2215
2216             For example, in `http://example.com/items`, the path is `/items`.
2217             """
2218         ),
2219     ],
2220     *,
2221     response_model: Annotated[
2222         Any,
2223         Doc(
2224             """
2225             The type to use for the response.
2226
2227             It could be any valid Pydantic *field* type. So, it doesn't have to
2228             be a Pydantic model, it could be other things, like a `list`, `dict`,
2229             etc.
2230
2231             It will be used for:
2232
2233             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
2234             show it as the response (JSON Schema).
2235             * Serialization: you could return an arbitrary object and the
2236             `response_model` would be used to serialize that object into the
2237             corresponding JSON.
2238             * Filtering: the JSON sent to the client will only contain the data
2239             (fields) defined in the `response_model`. If you returned an object
2240             that contains an attribute `password` but the `response_model` does
2241             not include that field, the JSON sent to the client would not have
2242             that `password`.
2243             * Validation: whatever you return will be serialized with the
2244             `response_model`, converting any data as necessary to generate the
2245             corresponding JSON. But if the data in the object returned is not
2246             valid, that would mean a violation of the contract with the client,
2247             so it's an error from the API developer. So, FastAPI will raise an
2248             error and return a 500 error code (Internal Server Error).
2249
2250             Read more about it in the
2251             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
2252             model/).
2253             """
2254         ),
2255     ] = Default(None),
2256     status_code: Annotated[
2257         int | None,
2258         Doc(
2259             """
2260             The default status code to be used for the response.
2261
2262             You could override the status code by returning a response directly.
2263
2264             Read more about it in the
2265             [FastAPI docs for Response Status Code]
2266             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
2267             """
2268         ),
2269     ] = None,
2270     tags: Annotated[
2271         list[str | Enum] | None,
2272         Doc(
2273             """
2274             A list of tags to be applied to the *path operation*.
2275
2276             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2277
2278             Read more about it in the
2279

```

```

2280         [FastAPI docs for Path Operation Configuration]
2281         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
2282         """
2283     ),
2284     ] = None,
2285     dependencies: Annotated[
2286         Sequence[params.Depends] | None,
2287         Doc(
2288             """
2289             A list of dependencies (using `Depends()`) to be applied to the
2290             *path operation*.
2291
2292             Read more about it in the
2293             [FastAPI docs for Dependencies in path operation decorators]
2294             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
2295             """
2296         ),
2297     ] = None,
2298     summary: Annotated[
2299         str | None,
2300         Doc(
2301             """
2302             A summary for the *path operation*.
2303
2304             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2305
2306             Read more about it in the
2307             [FastAPI docs for Path Operation Configuration]
2308             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2309             """
2310         ),
2311     ] = None,
2312     description: Annotated[
2313         str | None,
2314         Doc(
2315             """
2316             A description for the *path operation*.
2317
2318             If not provided, it will be extracted automatically from the docstring
2319             of the *path operation function*.
2320
2321             It can contain Markdown.
2322
2323             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2324
2325             Read more about it in the
2326             [FastAPI docs for Path Operation Configuration]
2327             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2328             """
2329         ),
2330     ] = None,
2331     response_description: Annotated[
2332         str,
2333         Doc(
2334             """
2335             The description for the default response.
2336
2337             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2338             """
2339         ),
2340     ] = "Successful Response",
2341     responses: Annotated[
2342         dict[int | str, dict[str, Any]] | None,
2343         Doc(
2344             """
2345             Additional responses that could be returned by this *path operation*.
2346
2347             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2348             """
2349         ),
2350     ] = None,
2351     deprecated: Annotated[
2352         bool | None,
2353         Doc(
2354             """
2355             Mark this *path operation* as deprecated.

```



```

2356         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2357         """
2358     ),
2359 ] = None,
2360 operation_id: Annotated[
2361     str | None,
2362     Doc(
2363         """
2364         Custom operation ID to be used by this *path operation*.
2365
2366         By default, it is generated automatically.
2367
2368         If you provide a custom operation ID, you need to make sure it is
2369         unique for the whole API.
2370
2371         You can customize the
2372         operation ID generation with the parameter
2373         `generate_unique_id_function` in the `FastAPI` class.
2374
2375         Read more about it in the
2376         [FastAPI docs about how to Generate Clients]
2377         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2378         """
2379     ),
2380 ] = None,
2381 response_model_include: Annotated[
2382     IncEx | None,
2383     Doc(
2384         """
2385         Configuration passed to Pydantic to include only certain fields in the
2386         response data.
2387
2388         Read more about it in the
2389         [FastAPI docs for Response Model - Return Type]
2390         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2391         """
2392     ),
2393 ] = None,
2394 response_model_exclude: Annotated[
2395     IncEx | None,
2396     Doc(
2397         """
2398         Configuration passed to Pydantic to exclude certain fields in the
2399         response data.
2400
2401         Read more about it in the
2402         [FastAPI docs for Response Model - Return Type]
2403         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2404         """
2405     ),
2406 ] = None,
2407 response_model_by_alias: Annotated[
2408     bool,
2409     Doc(
2410         """
2411         Configuration passed to Pydantic to define if the response model
2412         should be serialized by alias when an alias is used.
2413
2414         Read more about it in the
2415         [FastAPI docs for Response Model - Return Type]
2416         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2417         """
2418     ),
2419 ] = True,
2420 response_model_exclude_unset: Annotated[
2421     bool,
2422     Doc(
2423         """
2424         Configuration passed to Pydantic to define if the response data
2425         should have all the fields, including the ones that were not set and
2426         have their default values. This is different from
2427         `response_model_exclude_defaults` in that if the fields are set,
2428

```

```

2432         they will be included in the response, even if the value is the same
2433         as the default.
2434
2435         When `True`, default values are omitted from the response.
2436
2437         Read more about it in the
2438         [FastAPI docs for Response Model - Return Type]
2439         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2440         parameter).
2441         """
2442     ),
2443     ] = False,
2444     response_model_exclude_defaults: Annotated[
2445         bool,
2446         Doc(
2447             """
2448             Configuration passed to Pydantic to define if the response data
2449             should have all the fields, including the ones that have the same value
2450             as the default. This is different from `response_model_exclude_unset`
2451             in that if the fields are set but contain the same default values,
2452             they will be excluded from the response.
2453
2454             When `True`, default values are omitted from the response.
2455
2456             Read more about it in the
2457             [FastAPI docs for Response Model - Return Type]
2458             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2459             parameter).
2460             """
2461         ),
2462     ] = False,
2463     response_model_exclude_none: Annotated[
2464         bool,
2465         Doc(
2466             """
2467             Configuration passed to Pydantic to define if the response data should
2468             exclude fields set to `None`.
2469
2470             This is much simpler (less smart) than `response_model_exclude_unset`
2471             and `response_model_exclude_defaults`. You probably want to use one of
2472             those two instead of this one, as those allow returning `None` values
2473             when it makes sense.
2474
2475             Read more about it in the
2476             [FastAPI docs for Response Model - Return Type]
2477             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
2478             """
2479         ),
2480     ] = False,
2481     include_in_schema: Annotated[
2482         bool,
2483         Doc(
2484             """
2485             Include this *path operation* in the generated OpenAPI schema.
2486
2487             This affects the generated OpenAPI (e.g. visible at `/docs`).
2488
2489             Read more about it in the
2490             [FastAPI docs for Query Parameters and String Validations]
2491             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
2492             openapi).
2493             """
2494         ),
2495     ] = True,
2496     response_class: Annotated[
2497         type[Response],
2498         Doc(
2499             """
2500             Response class to be used for this *path operation*.
2501
2502             This will not be used if you return a response directly.
2503
2504             Read more about it in the
2505             [FastAPI docs for Custom Response - HTML, Stream, File, others]
2506             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
2507             """

```

```

2508     ),
2509     ] = Default(JSONResponse),
2510     name: Annotated[
2511         str | None,
2512         Doc(
2513             """
2514             Name for this *path operation*. Only used internally.
2515             """
2516         ),
2517     ] = None,
2518     callbacks: Annotated[
2519         list[BaseRoute] | None,
2520         Doc(
2521             """
2522             List of *path operations* that will be used as OpenAPI callbacks.
2523
2524             This is only for OpenAPI documentation, the callbacks won't be used
2525             directly.
2526
2527             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2528
2529             Read more about it in the
2530             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
2531 callbacks/).
2532             """
2533         ),
2534     ] = None,
2535     openapi_extra: Annotated[
2536         dict[str, Any] | None,
2537         Doc(
2538             """
2539             Extra metadata to be included in the OpenAPI schema for this *path
2540             operation*.
2541
2542             Read more about it in the
2543             [FastAPI docs for Path Operation Advanced Configuration]
2544             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
2545             path-operation-schema).
2546             """
2547         ),
2548     ] = None,
2549     generate_unique_id_function: Annotated[
2550         Callable[[APIRoute], str],
2551         Doc(
2552             """
2553             Customize the function used to generate unique IDs for the *path
2554             operations* shown in the generated OpenAPI.
2555
2556             This is particularly useful when automatically generating clients or
2557             SDKs for your API.
2558
2559             Read more about it in the
2560             [FastAPI docs about how to Generate Clients]
2561             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2562             """
2563         ),
2564     ] = Default(generate_unique_id),
2565 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
2566     """
2567     Add a *path operation* using an HTTP PUT operation.
2568
2569     ## Example
2570
2571     ```python
2572     from fastapi import APIRouter, FastAPI
2573     from pydantic import BaseModel
2574
2575     class Item(BaseModel):
2576         name: str
2577         description: str | None = None
2578
2579     app = FastAPI()
2580     router = APIRouter()
2581
2582     @router.put("/items/{item_id}")
2583     def replace_item(item_id: str, item: Item):

```

```

2584         return {"message": "Item replaced", "id": item_id}
2585
2586     app.include_router(router)
2587
2588     """
2589     return self.api_route(
2590         path=path,
2591         response_model=response_model,
2592         status_code=status_code,
2593         tags=tags,
2594         dependencies=dependencies,
2595         summary=summary,
2596         description=description,
2597         response_description=response_description,
2598         responses=responses,
2599         deprecated=deprecated,
2600         methods=["PUT"],
2601         operation_id=operation_id,
2602         response_model_include=response_model_include,
2603         response_model_exclude=response_model_exclude,
2604         response_model_by_alias=response_model_by_alias,
2605         response_model_exclude_unset=response_model_exclude_unset,
2606         response_model_exclude_defaults=response_model_exclude_defaults,
2607         response_model_exclude_none=response_model_exclude_none,
2608         include_in_schema=include_in_schema,
2609         response_class=response_class,
2610         name=name,
2611         callbacks=callbacks,
2612         openapi_extra=openapi_extra,
2613         generate_unique_id_function=generate_unique_id_function,
2614     )

```

meth post

```

post(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP POST operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI
from pydantic import BaseModel

class Item(BaseModel):

```

```
name: str
description: str | None = None

app = FastAPI()
router = APIRouter()

@router.post("/items/")
def create_item(item: Item):
    return {"message": "Item created"}

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: <code>Any</code> DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration.

PARAMETER	DESCRIPTION
	TYPE: <code>list[str Enum] None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients.

PARAMETER	DESCRIPTION
	TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

2590 def post(
2591     self,
2592     path: Annotated[
2593         str,
2594         Doc(
2595             """
2596             The URL path to be used for this *path* operation*.
2597
2598             For example, in `http://example.com/items`, the path is `/items`.
2599             """
2600         ),
2601     ],
2602     *,
2603     response_model: Annotated[
2604         Any,
2605         Doc(
2606             """
2607             The type to use for the response.
2608
2609             It could be any valid Pydantic *field* type. So, it doesn't have to
2610             be a Pydantic model, it could be other things, like a `list`, `dict`,
2611             etc.
2612
2613             It will be used for:
2614
2615             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
2616               show it as the response (JSON Schema).
2617             * Serialization: you could return an arbitrary object and the
2618               `response_model` would be used to serialize that object into the
2619               corresponding JSON.
2620             * Filtering: the JSON sent to the client will only contain the data
2621               (fields) defined in the `response_model`. If you returned an object
2622               that contains an attribute `password` but the `response_model` does
2623               not include that field, the JSON sent to the client would not have
2624               that `password`.
2625             * Validation: whatever you return will be serialized with the
2626               `response_model`, converting any data as necessary to generate the
2627               corresponding JSON. But if the data in the object returned is not
2628               valid, that would mean a violation of the contract with the client,
2629               so it's an error from the API developer. So, FastAPI will raise an
2630               error and return a 500 error code (Internal Server Error).
2631
2632               Read more about it in the
2633               [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
2634               model/).
2635             """
2636         ),
2637     ] = Default(None),
2638     status_code: Annotated[
2639         int | None,
2640         Doc(
2641             """
2642             The default status code to be used for the response.
2643
2644             You could override the status code by returning a response directly.
2645
2646             Read more about it in the
2647             [FastAPI docs for Response Status Code]
2648             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
2649             """
2650         ),
2651     ] = None,
2652     tags: Annotated[
2653         list[str | Enum] | None,
2654         Doc(
2655             """
2656             A list of tags to be applied to the *path* operation*.
2657
2658             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2659
2660             Read more about it in the
2661

```

```

2662         [FastAPI docs for Path Operation Configuration]
2663         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
2664         """
2665     ),
2666     ] = None,
2667     dependencies: Annotated[
2668         Sequence[params.Depends] | None,
2669         Doc(
2670             """
2671             A list of dependencies (using `Depends()`) to be applied to the
2672             *path operation*.
2673
2674             Read more about it in the
2675             [FastAPI docs for Dependencies in path operation decorators]
2676             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
2677             """
2678         ),
2679     ] = None,
2680     summary: Annotated[
2681         str | None,
2682         Doc(
2683             """
2684             A summary for the *path operation*.
2685
2686             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2687
2688             Read more about it in the
2689             [FastAPI docs for Path Operation Configuration]
2690             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2691             """
2692         ),
2693     ] = None,
2694     description: Annotated[
2695         str | None,
2696         Doc(
2697             """
2698             A description for the *path operation*.
2699
2700             If not provided, it will be extracted automatically from the docstring
2701             of the *path operation function*.
2702
2703             It can contain Markdown.
2704
2705             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2706
2707             Read more about it in the
2708             [FastAPI docs for Path Operation Configuration]
2709             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2710             """
2711         ),
2712     ] = None,
2713     response_description: Annotated[
2714         str,
2715         Doc(
2716             """
2717             The description for the default response.
2718
2719             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2720             """
2721         ),
2722     ] = "Successful Response",
2723     responses: Annotated[
2724         dict[int | str, dict[str, Any]] | None,
2725         Doc(
2726             """
2727             Additional responses that could be returned by this *path operation*.
2728
2729             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2730             """
2731         ),
2732     ] = None,
2733     deprecated: Annotated[
2734         bool | None,
2735         Doc(
2736             """
2737             Mark this *path operation* as deprecated.

```

```

2738         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2739         """
2740     ),
2741 ] = None,
2742 operation_id: Annotated[
2743     str | None,
2744     Doc(
2745         """
2746         Custom operation ID to be used by this *path operation*.
2747
2748         By default, it is generated automatically.
2749
2750         If you provide a custom operation ID, you need to make sure it is
2751         unique for the whole API.
2752
2753         You can customize the
2754         operation ID generation with the parameter
2755         `generate_unique_id_function` in the `FastAPI` class.
2756
2757         Read more about it in the
2758         [FastAPI docs about how to Generate Clients]
2759         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2760         """
2761     ),
2762 ] = None,
2763 response_model_include: Annotated[
2764     IncEx | None,
2765     Doc(
2766         """
2767         Configuration passed to Pydantic to include only certain fields in the
2768         response data.
2769
2770         Read more about it in the
2771         [FastAPI docs for Response Model - Return Type]
2772         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2773         """
2774     ),
2775 ] = None,
2776 response_model_exclude: Annotated[
2777     IncEx | None,
2778     Doc(
2779         """
2780         Configuration passed to Pydantic to exclude certain fields in the
2781         response data.
2782
2783         Read more about it in the
2784         [FastAPI docs for Response Model - Return Type]
2785         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2786         """
2787     ),
2788 ] = None,
2789 response_model_by_alias: Annotated[
2790     bool,
2791     Doc(
2792         """
2793         Configuration passed to Pydantic to define if the response model
2794         should be serialized by alias when an alias is used.
2795
2796         Read more about it in the
2797         [FastAPI docs for Response Model - Return Type]
2798         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2799         """
2800     ),
2801 ] = True,
2802 response_model_exclude_unset: Annotated[
2803     bool,
2804     Doc(
2805         """
2806         Configuration passed to Pydantic to define if the response data
2807         should have all the fields, including the ones that were not set and
2808         have their default values. This is different from
2809         `response_model_exclude_defaults` in that if the fields are set,
2810

```

```

2814         they will be included in the response, even if the value is the same
2815         as the default.
2816
2817         When `True`, default values are omitted from the response.
2818
2819         Read more about it in the
2820         [FastAPI docs for Response Model - Return Type]
2821         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2822         parameter).
2823         """
2824     ),
2825     ] = False,
2826     response_model_exclude_defaults: Annotated[
2827         bool,
2828         Doc(
2829             """
2830             Configuration passed to Pydantic to define if the response data
2831             should have all the fields, including the ones that have the same value
2832             as the default. This is different from `response_model_exclude_unset`
2833             in that if the fields are set but contain the same default values,
2834             they will be excluded from the response.
2835
2836             When `True`, default values are omitted from the response.
2837
2838             Read more about it in the
2839             [FastAPI docs for Response Model - Return Type]
2840             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2841             parameter).
2842             """
2843         ),
2844     ] = False,
2845     response_model_exclude_none: Annotated[
2846         bool,
2847         Doc(
2848             """
2849             Configuration passed to Pydantic to define if the response data should
2850             exclude fields set to `None`.
2851
2852             This is much simpler (less smart) than `response_model_exclude_unset`
2853             and `response_model_exclude_defaults`. You probably want to use one of
2854             those two instead of this one, as those allow returning `None` values
2855             when it makes sense.
2856
2857             Read more about it in the
2858             [FastAPI docs for Response Model - Return Type]
2859             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
2860             """
2861         ),
2862     ] = False,
2863     include_in_schema: Annotated[
2864         bool,
2865         Doc(
2866             """
2867             Include this *path operation* in the generated OpenAPI schema.
2868
2869             This affects the generated OpenAPI (e.g. visible at `/docs`).
2870
2871             Read more about it in the
2872             [FastAPI docs for Query Parameters and String Validations]
2873             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
2874             openapi).
2875             """
2876         ),
2877     ] = True,
2878     response_class: Annotated[
2879         type[Response],
2880         Doc(
2881             """
2882             Response class to be used for this *path operation*.
2883
2884             This will not be used if you return a response directly.
2885
2886             Read more about it in the
2887             [FastAPI docs for Custom Response - HTML, Stream, File, others]
2888             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
2889             """

```

```

2890     ),
2891     ] = Default(JSONResponse),
2892     name: Annotated[
2893         str | None,
2894         Doc(
2895             """
2896             Name for this *path operation*. Only used internally.
2897             """
2898         ),
2899     ] = None,
2900     callbacks: Annotated[
2901         list[BaseRoute] | None,
2902         Doc(
2903             """
2904             List of *path operations* that will be used as OpenAPI callbacks.
2905
2906             This is only for OpenAPI documentation, the callbacks won't be used
2907             directly.
2908
2909             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2910
2911             Read more about it in the
2912             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
2913             callbacks/).
2914             """
2915         ),
2916     ] = None,
2917     openapi_extra: Annotated[
2918         dict[str, Any] | None,
2919         Doc(
2920             """
2921             Extra metadata to be included in the OpenAPI schema for this *path
2922             operation*.
2923
2924             Read more about it in the
2925             [FastAPI docs for Path Operation Advanced Configuration]
2926             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
2927             path-operation-schema).
2928             """
2929         ),
2930     ] = None,
2931     generate_unique_id_function: Annotated[
2932         Callable[[APIRoute], str],
2933         Doc(
2934             """
2935             Customize the function used to generate unique IDs for the *path
2936             operations* shown in the generated OpenAPI.
2937
2938             This is particularly useful when automatically generating clients or
2939             SDKs for your API.
2940
2941             Read more about it in the
2942             [FastAPI docs about how to Generate Clients]
2943             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2944             """
2945         ),
2946     ] = Default(generate_unique_id),
2947 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
2948     """
2949     Add a *path operation* using an HTTP POST operation.
2950
2951     ## Example
2952
2953     ```python
2954     from fastapi import APIRouter, FastAPI
2955     from pydantic import BaseModel
2956
2957     class Item(BaseModel):
2958         name: str
2959         description: str | None = None
2960
2961     app = FastAPI()
2962     router = APIRouter()
2963
2964     @router.post("/items/")
2965     def create_item(item: Item):

```

```

2966         return {"message": "Item created"}
2967
2968     app.include_router(router)
2969     """
2970
2971     return self.api_route(
2972         path=path,
2973         response_model=response_model,
2974         status_code=status_code,
2975         tags=tags,
2976         dependencies=dependencies,
2977         summary=summary,
2978         description=description,
2979         response_description=response_description,
2980         responses=responses,
2981         deprecated=deprecated,
2982         methods=["POST"],
2983         operation_id=operation_id,
2984         response_model_include=response_model_include,
2985         response_model_exclude=response_model_exclude,
2986         response_model_by_alias=response_model_by_alias,
2987         response_model_exclude_unset=response_model_exclude_unset,
2988         response_model_exclude_defaults=response_model_exclude_defaults,
2989         response_model_exclude_none=response_model_exclude_none,
2990         include_in_schema=include_in_schema,
2991         response_class=response_class,
2992         name=name,
2993         callbacks=callbacks,
2994         openapi_extra=openapi_extra,
2995         generate_unique_id_function=generate_unique_id_function,
2996     )

```

meth delete

```

delete(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP DELETE operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI

app = FastAPI()
router = APIRouter()

```

```
@router.delete("/items/{item_id}")
def delete_item(item_id: str):
    return {"message": "Item deleted"}

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> <code>None</code> DEFAULT: <code>None</code>
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

2972 def delete(
2973     self,
2974     path: Annotated[
2975         str,
2976         Doc(
2977             """
2978             The URL path to be used for this *path operation*.
2979
2980             For example, in `http://example.com/items`, the path is `/items`.
2981             """
2982         ),
2983     ],
2984     *,
2985     response_model: Annotated[
2986         Any,
2987         Doc(
2988             """
2989             The type to use for the response.
2990
2991             It could be any valid Pydantic *field* type. So, it doesn't have to
2992             be a Pydantic model, it could be other things, like a `list`, `dict`,
2993             etc.
2994
2995             It will be used for:
2996
2997             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
2998               show it as the response (JSON Schema).
2999             * Serialization: you could return an arbitrary object and the
3000               `response_model` would be used to serialize that object into the
3001               corresponding JSON.
3002             * Filtering: the JSON sent to the client will only contain the data
3003               (fields) defined in the `response_model`. If you returned an object
3004               that contains an attribute `password` but the `response_model` does
3005               not include that field, the JSON sent to the client would not have
3006               that `password`.
3007             * Validation: whatever you return will be serialized with the
3008               `response_model`, converting any data as necessary to generate the
3009               corresponding JSON. But if the data in the object returned is not
3010               valid, that would mean a violation of the contract with the client,
3011               so it's an error from the API developer. So, FastAPI will raise an
3012               error and return a 500 error code (Internal Server Error).
3013
3014             Read more about it in the
3015             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
3016             model/).
3017             """
3018         ),
3019     ] = Default(None),
3020     status_code: Annotated[
3021         int | None,
3022         Doc(
3023             """
3024             The default status code to be used for the response.
3025
3026             You could override the status code by returning a response directly.
3027
3028             Read more about it in the
3029             [FastAPI docs for Response Status Code]
3030             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
3031             """
3032         ),
3033     ] = None,
3034     tags: Annotated[
3035         list[str | Enum] | None,
3036         Doc(
3037             """
3038             A list of tags to be applied to the *path operation*.
3039
3040             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3041
3042             Read more about it in the
3043             """

```

```

3044         [FastAPI docs for Path Operation Configuration]
3045         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
3046         """
3047     ),
3048     ] = None,
3049     dependencies: Annotated[
3050         Sequence[params.Depends] | None,
3051         Doc(
3052             """
3053             A list of dependencies (using `Depends()`) to be applied to the
3054             *path operation*.
3055
3056             Read more about it in the
3057             [FastAPI docs for Dependencies in path operation decorators]
3058             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
3059             """
3060         ),
3061     ] = None,
3062     summary: Annotated[
3063         str | None,
3064         Doc(
3065             """
3066             A summary for the *path operation*.
3067
3068             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3069
3070             Read more about it in the
3071             [FastAPI docs for Path Operation Configuration]
3072             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3073             """
3074         ),
3075     ] = None,
3076     description: Annotated[
3077         str | None,
3078         Doc(
3079             """
3080             A description for the *path operation*.
3081
3082             If not provided, it will be extracted automatically from the docstring
3083             of the *path operation function*.
3084
3085             It can contain Markdown.
3086
3087             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3088
3089             Read more about it in the
3090             [FastAPI docs for Path Operation Configuration]
3091             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3092             """
3093         ),
3094     ] = None,
3095     response_description: Annotated[
3096         str,
3097         Doc(
3098             """
3099             The description for the default response.
3100
3101             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3102             """
3103         ),
3104     ] = "Successful Response",
3105     responses: Annotated[
3106         dict[int | str, dict[str, Any]] | None,
3107         Doc(
3108             """
3109             Additional responses that could be returned by this *path operation*.
3110
3111             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3112             """
3113         ),
3114     ] = None,
3115     deprecated: Annotated[
3116         bool | None,
3117         Doc(
3118             """
3119             Mark this *path operation* as deprecated.

```

```

3120         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3121         """
3122     ),
3123 ] = None,
3124 operation_id: Annotated[
3125     str | None,
3126     Doc(
3127         """
3128         Custom operation ID to be used by this *path operation*.
3129
3130         By default, it is generated automatically.
3131
3132         If you provide a custom operation ID, you need to make sure it is
3133         unique for the whole API.
3134
3135         You can customize the
3136         operation ID generation with the parameter
3137         `generate_unique_id_function` in the `FastAPI` class.
3138
3139         Read more about it in the
3140         [FastAPI docs about how to Generate Clients]
3141         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3142         """
3143     ),
3144 ] = None,
3145 response_model_include: Annotated[
3146     IncEx | None,
3147     Doc(
3148         """
3149         Configuration passed to Pydantic to include only certain fields in the
3150         response data.
3151
3152         Read more about it in the
3153         [FastAPI docs for Response Model - Return Type]
3154         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3155         """
3156     ),
3157 ] = None,
3158 response_model_exclude: Annotated[
3159     IncEx | None,
3160     Doc(
3161         """
3162         Configuration passed to Pydantic to exclude certain fields in the
3163         response data.
3164
3165         Read more about it in the
3166         [FastAPI docs for Response Model - Return Type]
3167         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3168         """
3169     ),
3170 ] = None,
3171 response_model_by_alias: Annotated[
3172     bool,
3173     Doc(
3174         """
3175         Configuration passed to Pydantic to define if the response model
3176         should be serialized by alias when an alias is used.
3177
3178         Read more about it in the
3179         [FastAPI docs for Response Model - Return Type]
3180         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3181         """
3182     ),
3183 ] = True,
3184 response_model_exclude_unset: Annotated[
3185     bool,
3186     Doc(
3187         """
3188         Configuration passed to Pydantic to define if the response data
3189         should have all the fields, including the ones that were not set and
3190         have their default values. This is different from
3191         `response_model_exclude_defaults` in that if the fields are set,

```

```

3196         they will be included in the response, even if the value is the same
3197         as the default.
3198
3199         When `True`, default values are omitted from the response.
3200
3201         Read more about it in the
3202         [FastAPI docs for Response Model - Return Type]
3203         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3204         parameter).
3205         """
3206     ),
3207     ] = False,
3208     response_model_exclude_defaults: Annotated[
3209         bool,
3210         Doc(
3211             """
3212             Configuration passed to Pydantic to define if the response data
3213             should have all the fields, including the ones that have the same value
3214             as the default. This is different from `response_model_exclude_unset`
3215             in that if the fields are set but contain the same default values,
3216             they will be excluded from the response.
3217
3218             When `True`, default values are omitted from the response.
3219
3220             Read more about it in the
3221             [FastAPI docs for Response Model - Return Type]
3222             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3223             parameter).
3224             """
3225         ),
3226     ] = False,
3227     response_model_exclude_none: Annotated[
3228         bool,
3229         Doc(
3230             """
3231             Configuration passed to Pydantic to define if the response data should
3232             exclude fields set to `None`.
3233
3234             This is much simpler (less smart) than `response_model_exclude_unset`
3235             and `response_model_exclude_defaults`. You probably want to use one of
3236             those two instead of this one, as those allow returning `None` values
3237             when it makes sense.
3238
3239             Read more about it in the
3240             [FastAPI docs for Response Model - Return Type]
3241             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
3242             """
3243         ),
3244     ] = False,
3245     include_in_schema: Annotated[
3246         bool,
3247         Doc(
3248             """
3249             Include this *path operation* in the generated OpenAPI schema.
3250
3251             This affects the generated OpenAPI (e.g. visible at `/docs`).
3252
3253             Read more about it in the
3254             [FastAPI docs for Query Parameters and String Validations]
3255             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
3256             openapi).
3257             """
3258         ),
3259     ] = True,
3260     response_class: Annotated[
3261         type[Response],
3262         Doc(
3263             """
3264             Response class to be used for this *path operation*.
3265
3266             This will not be used if you return a response directly.
3267
3268             Read more about it in the
3269             [FastAPI docs for Custom Response - HTML, Stream, File, others]
3270             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
3271             """

```

```

3272     ),
3273     ] = Default(JSONResponse),
3274     name: Annotated[
3275         str | None,
3276         Doc(
3277             """
3278             Name for this *path operation*. Only used internally.
3279             """
3280         ),
3281     ] = None,
3282     callbacks: Annotated[
3283         list[BaseRoute] | None,
3284         Doc(
3285             """
3286             List of *path operations* that will be used as OpenAPI callbacks.
3287
3288             This is only for OpenAPI documentation, the callbacks won't be used
3289             directly.
3290
3291             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3292
3293             Read more about it in the
3294             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
3295             callbacks/).
3296             """
3297         ),
3298     ] = None,
3299     openapi_extra: Annotated[
3300         dict[str, Any] | None,
3301         Doc(
3302             """
3303             Extra metadata to be included in the OpenAPI schema for this *path
3304             operation*.
3305
3306             Read more about it in the
3307             [FastAPI docs for Path Operation Advanced Configuration]
3308             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
3309             path-operation-schema).
3310             """
3311         ),
3312     ] = None,
3313     generate_unique_id_function: Annotated[
3314         Callable[[APIRoute], str],
3315         Doc(
3316             """
3317             Customize the function used to generate unique IDs for the *path
3318             operations* shown in the generated OpenAPI.
3319
3320             This is particularly useful when automatically generating clients or
3321             SDKs for your API.
3322
3323             Read more about it in the
3324             [FastAPI docs about how to Generate Clients]
3325             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3326             """
3327         ),
3328     ] = Default(generate_unique_id),
3329 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
3330     """
3331     Add a *path operation* using an HTTP DELETE operation.
3332
3333     ## Example
3334
3335     ```python
3336     from fastapi import APIRouter, FastAPI
3337
3338     app = FastAPI()
3339     router = APIRouter()
3340
3341     @router.delete("/items/{item_id}")
3342     def delete_item(item_id: str):
3343         return {"message": "Item deleted"}
3344
3345     app.include_router(router)
3346     """
3347

```



```

    return self.api_route(
        path=path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        methods=["DELETE"],
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth options

```

options(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP OPTIONS operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI

app = FastAPI()
router = APIRouter()

@router.options("/items/")
def get_item_options():
    return {"additions": ["Aji", "Guacamole"]}

```

```
app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: <code>Any</code> DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> <code>None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

3349 def options(
3350     self,
3351     path: Annotated[
3352         str,
3353         Doc(
3354             """
3355             The URL path to be used for this *path* operation*.
3356
3357             For example, in `http://example.com/items`, the path is `/items`.
3358             """
3359         ),
3360     ],
3361     *,
3362     response_model: Annotated[
3363         Any,
3364         Doc(
3365             """
3366             The type to use for the response.
3367
3368             It could be any valid Pydantic *field* type. So, it doesn't have to
3369             be a Pydantic model, it could be other things, like a `list`, `dict`,
3370             etc.
3371
3372             It will be used for:
3373
3374             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
3375               show it as the response (JSON Schema).
3376             * Serialization: you could return an arbitrary object and the
3377               `response_model` would be used to serialize that object into the
3378               corresponding JSON.
3379             * Filtering: the JSON sent to the client will only contain the data
3380               (fields) defined in the `response_model`. If you returned an object
3381               that contains an attribute `password` but the `response_model` does
3382               not include that field, the JSON sent to the client would not have
3383               that `password`.
3384             * Validation: whatever you return will be serialized with the
3385               `response_model`, converting any data as necessary to generate the
3386               corresponding JSON. But if the data in the object returned is not
3387               valid, that would mean a violation of the contract with the client,
3388               so it's an error from the API developer. So, FastAPI will raise an
3389               error and return a 500 error code (Internal Server Error).
3390
3391             Read more about it in the
3392             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
3393             model/).
3394             """
3395         ),
3396     ] = Default(None),
3397     status_code: Annotated[
3398         int | None,
3399         Doc(
3400             """
3401             The default status code to be used for the response.
3402
3403             You could override the status code by returning a response directly.
3404
3405             Read more about it in the
3406             [FastAPI docs for Response Status Code]
3407             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
3408             """
3409         ),
3410     ] = None,
3411     tags: Annotated[
3412         list[str | Enum] | None,
3413         Doc(
3414             """
3415             A list of tags to be applied to the *path* operation*.
3416
3417             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3418
3419             Read more about it in the
3420

```

```

3421         [FastAPI docs for Path Operation Configuration]
3422         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
3423         """
3424     ),
3425     ] = None,
3426     dependencies: Annotated[
3427         Sequence[params.Depends] | None,
3428         Doc(
3429             """
3430             A list of dependencies (using `Depends()`) to be applied to the
3431             *path operation*.
3432
3433             Read more about it in the
3434             [FastAPI docs for Dependencies in path operation decorators]
3435             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
3436             """
3437         ),
3438     ] = None,
3439     summary: Annotated[
3440         str | None,
3441         Doc(
3442             """
3443             A summary for the *path operation*.
3444
3445             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3446
3447             Read more about it in the
3448             [FastAPI docs for Path Operation Configuration]
3449             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3450             """
3451         ),
3452     ] = None,
3453     description: Annotated[
3454         str | None,
3455         Doc(
3456             """
3457             A description for the *path operation*.
3458
3459             If not provided, it will be extracted automatically from the docstring
3460             of the *path operation* function*.
3461
3462             It can contain Markdown.
3463
3464             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3465
3466             Read more about it in the
3467             [FastAPI docs for Path Operation Configuration]
3468             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3469             """
3470         ),
3471     ] = None,
3472     response_description: Annotated[
3473         str,
3474         Doc(
3475             """
3476             The description for the default response.
3477
3478             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3479             """
3480         ),
3481     ] = "Successful Response",
3482     responses: Annotated[
3483         dict[int | str, dict[str, Any]] | None,
3484         Doc(
3485             """
3486             Additional responses that could be returned by this *path operation*.
3487
3488             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3489             """
3490         ),
3491     ] = None,
3492     deprecated: Annotated[
3493         bool | None,
3494         Doc(
3495             """
3496             Mark this *path operation* as deprecated.

```

```

3497         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3498         """
3499     ),
3500 ] = None,
3501 operation_id: Annotated[
3502     str | None,
3503     Doc(
3504         """
3505         Custom operation ID to be used by this *path operation*.
3506
3507         By default, it is generated automatically.
3508
3509         If you provide a custom operation ID, you need to make sure it is
3510         unique for the whole API.
3511
3512         You can customize the
3513         operation ID generation with the parameter
3514         `generate_unique_id_function` in the `FastAPI` class.
3515
3516         Read more about it in the
3517         [FastAPI docs about how to Generate Clients]
3518         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3519         """
3520     ),
3521 ] = None,
3522 response_model_include: Annotated[
3523     IncEx | None,
3524     Doc(
3525         """
3526         Configuration passed to Pydantic to include only certain fields in the
3527         response data.
3528
3529         Read more about it in the
3530         [FastAPI docs for Response Model - Return Type]
3531         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3532         """
3533     ),
3534 ] = None,
3535 response_model_exclude: Annotated[
3536     IncEx | None,
3537     Doc(
3538         """
3539         Configuration passed to Pydantic to exclude certain fields in the
3540         response data.
3541
3542         Read more about it in the
3543         [FastAPI docs for Response Model - Return Type]
3544         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3545         """
3546     ),
3547 ] = None,
3548 response_model_by_alias: Annotated[
3549     bool,
3550     Doc(
3551         """
3552         Configuration passed to Pydantic to define if the response model
3553         should be serialized by alias when an alias is used.
3554
3555         Read more about it in the
3556         [FastAPI docs for Response Model - Return Type]
3557         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3558         """
3559     ),
3560 ] = True,
3561 response_model_exclude_unset: Annotated[
3562     bool,
3563     Doc(
3564         """
3565         Configuration passed to Pydantic to define if the response data
3566         should have all the fields, including the ones that were not set and
3567         have their default values. This is different from
3568         `response_model_exclude_defaults` in that if the fields are set,

```



```

3573         they will be included in the response, even if the value is the same
3574         as the default.
3575
3576         When `True`, default values are omitted from the response.
3577
3578         Read more about it in the
3579         [FastAPI docs for Response Model - Return Type]
3580         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3581         parameter).
3582         """
3583     ),
3584     ] = False,
3585     response_model_exclude_defaults: Annotated[
3586         bool,
3587         Doc(
3588             """
3589             Configuration passed to Pydantic to define if the response data
3590             should have all the fields, including the ones that have the same value
3591             as the default. This is different from `response_model_exclude_unset`
3592             in that if the fields are set but contain the same default values,
3593             they will be excluded from the response.
3594
3595             When `True`, default values are omitted from the response.
3596
3597             Read more about it in the
3598             [FastAPI docs for Response Model - Return Type]
3599             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3600             parameter).
3601             """
3602         ),
3603     ] = False,
3604     response_model_exclude_none: Annotated[
3605         bool,
3606         Doc(
3607             """
3608             Configuration passed to Pydantic to define if the response data should
3609             exclude fields set to `None`.
3610
3611             This is much simpler (less smart) than `response_model_exclude_unset`
3612             and `response_model_exclude_defaults`. You probably want to use one of
3613             those two instead of this one, as those allow returning `None` values
3614             when it makes sense.
3615
3616             Read more about it in the
3617             [FastAPI docs for Response Model - Return Type]
3618             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
3619             """
3620         ),
3621     ] = False,
3622     include_in_schema: Annotated[
3623         bool,
3624         Doc(
3625             """
3626             Include this *path operation* in the generated OpenAPI schema.
3627
3628             This affects the generated OpenAPI (e.g. visible at `/docs`).
3629
3630             Read more about it in the
3631             [FastAPI docs for Query Parameters and String Validations]
3632             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
3633             openapi).
3634             """
3635         ),
3636     ] = True,
3637     response_class: Annotated[
3638         type[Response],
3639         Doc(
3640             """
3641             Response class to be used for this *path operation*.
3642
3643             This will not be used if you return a response directly.
3644
3645             Read more about it in the
3646             [FastAPI docs for Custom Response - HTML, Stream, File, others]
3647             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
3648             """

```

```

3649     ),
3650     ] = Default(JSONResponse),
3651     name: Annotated[
3652         str | None,
3653         Doc(
3654             """
3655             Name for this *path operation*. Only used internally.
3656             """
3657         ),
3658     ] = None,
3659     callbacks: Annotated[
3660         list[BaseRoute] | None,
3661         Doc(
3662             """
3663             List of *path operations* that will be used as OpenAPI callbacks.
3664
3665             This is only for OpenAPI documentation, the callbacks won't be used
3666             directly.
3667
3668             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3669
3670             Read more about it in the
3671             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
3672             callbacks/).
3673             """
3674         ),
3675     ] = None,
3676     openapi_extra: Annotated[
3677         dict[str, Any] | None,
3678         Doc(
3679             """
3680             Extra metadata to be included in the OpenAPI schema for this *path
3681             operation*.
3682
3683             Read more about it in the
3684             [FastAPI docs for Path Operation Advanced Configuration]
3685             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
3686             path-operation-schema).
3687             """
3688         ),
3689     ] = None,
3690     generate_unique_id_function: Annotated[
3691         Callable[[APIRoute], str],
3692         Doc(
3693             """
3694             Customize the function used to generate unique IDs for the *path
3695             operations* shown in the generated OpenAPI.
3696
3697             This is particularly useful when automatically generating clients or
3698             SDKs for your API.
3699
3700             Read more about it in the
3701             [FastAPI docs about how to Generate Clients]
3702             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3703             """
3704         ),
3705     ] = Default(generate_unique_id),
3706 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
3707     """
3708     Add a *path operation* using an HTTP OPTIONS operation.
3709
3710     ## Example
3711
3712     ```python
3713     from fastapi import APIRouter, FastAPI
3714
3715     app = FastAPI()
3716     router = APIRouter()
3717
3718     @router.options("/items/")
3719     def get_item_options():
3720         return {"additions": ["Aji", "Guacamole"]}
3721
3722     app.include_router(router)
3723     """
3724

```

```

    return self.api_route(
        path=path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        methods=["OPTIONS"],
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth head

```

head(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP HEAD operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI
from pydantic import BaseModel

class Item(BaseModel):
    name: str
    description: str | None = None

app = FastAPI()
router = APIRouter()

```

```
@router.head("/items/", status_code=204)
def get_items_headers(response: Response):
    response.headers["X-Cat-Dog"] = "Alone in the world"

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> None DEFAULT: None
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> None DEFAULT: None

PARAMETER	DESCRIPTION
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> <code>None</code> DEFAULT: <code>None</code>
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

3726 def head(
3727     self,
3728     path: Annotated[
3729         str,
3730         Doc(
3731             """
3732             The URL path to be used for this *path operation*.
3733
3734             For example, in `http://example.com/items`, the path is `/items`.
3735             """
3736         ),
3737     ],
3738     *,
3739     response_model: Annotated[
3740         Any,
3741         Doc(
3742             """
3743             The type to use for the response.
3744
3745             It could be any valid Pydantic *field* type. So, it doesn't have to
3746             be a Pydantic model, it could be other things, like a `list`, `dict`,
3747             etc.
3748
3749             It will be used for:
3750
3751             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
3752               show it as the response (JSON Schema).
3753             * Serialization: you could return an arbitrary object and the
3754               `response_model` would be used to serialize that object into the
3755               corresponding JSON.
3756             * Filtering: the JSON sent to the client will only contain the data
3757               (fields) defined in the `response_model`. If you returned an object
3758               that contains an attribute `password` but the `response_model` does
3759               not include that field, the JSON sent to the client would not have
3760               that `password`.
3761             * Validation: whatever you return will be serialized with the
3762               `response_model`, converting any data as necessary to generate the
3763               corresponding JSON. But if the data in the object returned is not
3764               valid, that would mean a violation of the contract with the client,
3765               so it's an error from the API developer. So, FastAPI will raise an
3766               error and return a 500 error code (Internal Server Error).
3767
3768             Read more about it in the
3769             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
3770             model/).
3771             """
3772         ),
3773     ] = Default(None),
3774     status_code: Annotated[
3775         int | None,
3776         Doc(
3777             """
3778             The default status code to be used for the response.
3779
3780             You could override the status code by returning a response directly.
3781
3782             Read more about it in the
3783             [FastAPI docs for Response Status Code]
3784             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
3785             """
3786         ),
3787     ] = None,
3788     tags: Annotated[
3789         list[str | Enum] | None,
3790         Doc(
3791             """
3792             A list of tags to be applied to the *path operation*.
3793
3794             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3795
3796             Read more about it in the
3797

```



```

3798         [FastAPI docs for Path Operation Configuration]
3799         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
3800         """
3801     ),
3802     ] = None,
3803     dependencies: Annotated[
3804         Sequence[params.Depends] | None,
3805         Doc(
3806             """
3807             A list of dependencies (using `Depends()`) to be applied to the
3808             *path operation*.
3809
3810             Read more about it in the
3811             [FastAPI docs for Dependencies in path operation decorators]
3812             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
3813             """
3814         ),
3815     ] = None,
3816     summary: Annotated[
3817         str | None,
3818         Doc(
3819             """
3820             A summary for the *path operation*.
3821
3822             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3823
3824             Read more about it in the
3825             [FastAPI docs for Path Operation Configuration]
3826             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3827             """
3828         ),
3829     ] = None,
3830     description: Annotated[
3831         str | None,
3832         Doc(
3833             """
3834             A description for the *path operation*.
3835
3836             If not provided, it will be extracted automatically from the docstring
3837             of the *path operation function*.
3838
3839             It can contain Markdown.
3840
3841             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3842
3843             Read more about it in the
3844             [FastAPI docs for Path Operation Configuration]
3845             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3846             """
3847         ),
3848     ] = None,
3849     response_description: Annotated[
3850         str,
3851         Doc(
3852             """
3853             The description for the default response.
3854
3855             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3856             """
3857         ),
3858     ] = "Successful Response",
3859     responses: Annotated[
3860         dict[int | str, dict[str, Any]] | None,
3861         Doc(
3862             """
3863             Additional responses that could be returned by this *path operation*.
3864
3865             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3866             """
3867         ),
3868     ] = None,
3869     deprecated: Annotated[
3870         bool | None,
3871         Doc(
3872             """
3873             Mark this *path operation* as deprecated.

```

```

3874         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3875         """
3876     ),
3877 ] = None,
3878 operation_id: Annotated[
3879     str | None,
3880     Doc(
3881         """
3882         Custom operation ID to be used by this *path operation*.
3883
3884         By default, it is generated automatically.
3885
3886         If you provide a custom operation ID, you need to make sure it is
3887         unique for the whole API.
3888
3889         You can customize the
3890         operation ID generation with the parameter
3891         `generate_unique_id_function` in the `FastAPI` class.
3892
3893         Read more about it in the
3894         [FastAPI docs about how to Generate Clients]
3895         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3896         """
3897     ),
3898 ] = None,
3899 response_model_include: Annotated[
3900     IncEx | None,
3901     Doc(
3902         """
3903         Configuration passed to Pydantic to include only certain fields in the
3904         response data.
3905
3906         Read more about it in the
3907         [FastAPI docs for Response Model - Return Type]
3908         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3909         """
3910     ),
3911 ] = None,
3912 response_model_exclude: Annotated[
3913     IncEx | None,
3914     Doc(
3915         """
3916         Configuration passed to Pydantic to exclude certain fields in the
3917         response data.
3918
3919         Read more about it in the
3920         [FastAPI docs for Response Model - Return Type]
3921         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3922         """
3923     ),
3924 ] = None,
3925 response_model_by_alias: Annotated[
3926     bool,
3927     Doc(
3928         """
3929         Configuration passed to Pydantic to define if the response model
3930         should be serialized by alias when an alias is used.
3931
3932         Read more about it in the
3933         [FastAPI docs for Response Model - Return Type]
3934         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3935         """
3936     ),
3937 ] = True,
3938 response_model_exclude_unset: Annotated[
3939     bool,
3940     Doc(
3941         """
3942         Configuration passed to Pydantic to define if the response data
3943         should have all the fields, including the ones that were not set and
3944         have their default values. This is different from
3945         `response_model_exclude_defaults` in that if the fields are set,

```

```

3950         they will be included in the response, even if the value is the same
3951         as the default.
3952
3953         When `True`, default values are omitted from the response.
3954
3955         Read more about it in the
3956         [FastAPI docs for Response Model - Return Type]
3957         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3958         parameter).
3959         """
3960     ),
3961     ] = False,
3962     response_model_exclude_defaults: Annotated[
3963         bool,
3964         Doc(
3965             """
3966             Configuration passed to Pydantic to define if the response data
3967             should have all the fields, including the ones that have the same value
3968             as the default. This is different from `response_model_exclude_unset`
3969             in that if the fields are set but contain the same default values,
3970             they will be excluded from the response.
3971
3972             When `True`, default values are omitted from the response.
3973
3974             Read more about it in the
3975             [FastAPI docs for Response Model - Return Type]
3976             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3977             parameter).
3978             """
3979         ),
3980     ] = False,
3981     response_model_exclude_none: Annotated[
3982         bool,
3983         Doc(
3984             """
3985             Configuration passed to Pydantic to define if the response data should
3986             exclude fields set to `None`.
3987
3988             This is much simpler (less smart) than `response_model_exclude_unset`
3989             and `response_model_exclude_defaults`. You probably want to use one of
3990             those two instead of this one, as those allow returning `None` values
3991             when it makes sense.
3992
3993             Read more about it in the
3994             [FastAPI docs for Response Model - Return Type]
3995             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
3996             """
3997         ),
3998     ] = False,
3999     include_in_schema: Annotated[
4000         bool,
4001         Doc(
4002             """
4003             Include this *path operation* in the generated OpenAPI schema.
4004
4005             This affects the generated OpenAPI (e.g. visible at `/docs`).
4006
4007             Read more about it in the
4008             [FastAPI docs for Query Parameters and String Validations]
4009             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
4010             openapi).
4011             """
4012         ),
4013     ] = True,
4014     response_class: Annotated[
4015         type[Response],
4016         Doc(
4017             """
4018             Response class to be used for this *path operation*.
4019
4020             This will not be used if you return a response directly.
4021
4022             Read more about it in the
4023             [FastAPI docs for Custom Response - HTML, Stream, File, others]
4024             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
4025             """

```

```

4026     ),
4027     ] = Default(JSONResponse),
4028     name: Annotated[
4029         str | None,
4030         Doc(
4031             """
4032             Name for this *path operation*. Only used internally.
4033             """
4034         ),
4035     ] = None,
4036     callbacks: Annotated[
4037         list[BaseRoute] | None,
4038         Doc(
4039             """
4040             List of *path operations* that will be used as OpenAPI callbacks.
4041
4042             This is only for OpenAPI documentation, the callbacks won't be used
4043             directly.
4044
4045             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4046
4047             Read more about it in the
4048             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
4049             callbacks/).
4050             """
4051         ),
4052     ] = None,
4053     openapi_extra: Annotated[
4054         dict[str, Any] | None,
4055         Doc(
4056             """
4057             Extra metadata to be included in the OpenAPI schema for this *path
4058             operation*.
4059
4060             Read more about it in the
4061             [FastAPI docs for Path Operation Advanced Configuration]
4062             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
4063             path-operation-schema).
4064             """
4065         ),
4066     ] = None,
4067     generate_unique_id_function: Annotated[
4068         Callable[[APIRoute], str],
4069         Doc(
4070             """
4071             Customize the function used to generate unique IDs for the *path
4072             operations* shown in the generated OpenAPI.
4073
4074             This is particularly useful when automatically generating clients or
4075             SDKs for your API.
4076
4077             Read more about it in the
4078             [FastAPI docs about how to Generate Clients]
4079             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4080             """
4081         ),
4082     ] = Default(generate_unique_id),
4083 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4084     """
4085     Add a *path operation* using an HTTP HEAD operation.
4086
4087     ## Example
4088
4089     ```python
4090     from fastapi import APIRouter, FastAPI
4091     from pydantic import BaseModel
4092
4093     class Item(BaseModel):
4094         name: str
4095         description: str | None = None
4096
4097     app = FastAPI()
4098     router = APIRouter()
4099
4100     @router.head("/items/", status_code=204)
4101     def get_items_headers(response: Response):

```

```

4102         response.headers["X-Cat-Dog"] = "Alone in the world"
4103
4104     app.include_router(router)
4105     """
4106     return self.api_route(
        path=path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        methods=["HEAD"],
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth patch

```

patch(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP PATCH operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI
from pydantic import BaseModel

class Item(BaseModel):

```

```
name: str
description: str | None = None

app = FastAPI()
router = APIRouter()

@router.patch("/items/")
def update_item(item: Item):
    return {"message": "Item updated in place"}

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: <code>Any</code> DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration.

PARAMETER	DESCRIPTION		
		TYPE:	DEFAULT:
		<code>list[str Enum] None</code>	<code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>		
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>		
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>		
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>		
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>		
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool None</code> DEFAULT: <code>None</code>		

PARAMETER	DESCRIPTION
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients.

PARAMETER	DESCRIPTION
	TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

4108 def patch(
4109     self,
4110     path: Annotated[
4111         str,
4112         Doc(
4113             """
4114             The URL path to be used for this *path* operation*.
4115
4116             For example, in `http://example.com/items`, the path is `/items`.
4117             """
4118         ),
4119     ],
4120     *,
4121     response_model: Annotated[
4122         Any,
4123         Doc(
4124             """
4125             The type to use for the response.
4126
4127             It could be any valid Pydantic *field* type. So, it doesn't have to
4128             be a Pydantic model, it could be other things, like a `list`, `dict`,
4129             etc.
4130
4131             It will be used for:
4132
4133             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
4134               show it as the response (JSON Schema).
4135             * Serialization: you could return an arbitrary object and the
4136               `response_model` would be used to serialize that object into the
4137               corresponding JSON.
4138             * Filtering: the JSON sent to the client will only contain the data
4139               (fields) defined in the `response_model`. If you returned an object
4140               that contains an attribute `password` but the `response_model` does
4141               not include that field, the JSON sent to the client would not have
4142               that `password`.
4143             * Validation: whatever you return will be serialized with the
4144               `response_model`, converting any data as necessary to generate the
4145               corresponding JSON. But if the data in the object returned is not
4146               valid, that would mean a violation of the contract with the client,
4147               so it's an error from the API developer. So, FastAPI will raise an
4148               error and return a 500 error code (Internal Server Error).
4149
4150             Read more about it in the
4151             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
4152             model/).
4153             """
4154         ),
4155     ] = Default(None),
4156     status_code: Annotated[
4157         int | None,
4158         Doc(
4159             """
4160             The default status code to be used for the response.
4161
4162             You could override the status code by returning a response directly.
4163
4164             Read more about it in the
4165             [FastAPI docs for Response Status Code]
4166             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
4167             """
4168         ),
4169     ] = None,
4170     tags: Annotated[
4171         list[str | Enum] | None,
4172         Doc(
4173             """
4174             A list of tags to be applied to the *path* operation*.
4175
4176             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4177
4178             Read more about it in the
4179

```

```

4180         [FastAPI docs for Path Operation Configuration]
4181         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
4182         """
4183     ),
4184     ] = None,
4185     dependencies: Annotated[
4186         Sequence[params.Depends] | None,
4187         Doc(
4188             """
4189             A list of dependencies (using `Depends()`) to be applied to the
4190             *path operation*.
4191
4192             Read more about it in the
4193             [FastAPI docs for Dependencies in path operation decorators]
4194             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
4195             """
4196         ),
4197     ] = None,
4198     summary: Annotated[
4199         str | None,
4200         Doc(
4201             """
4202             A summary for the *path operation*.
4203
4204             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4205
4206             Read more about it in the
4207             [FastAPI docs for Path Operation Configuration]
4208             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
4209             """
4210         ),
4211     ] = None,
4212     description: Annotated[
4213         str | None,
4214         Doc(
4215             """
4216             A description for the *path operation*.
4217
4218             If not provided, it will be extracted automatically from the docstring
4219             of the *path operation function*.
4220
4221             It can contain Markdown.
4222
4223             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4224
4225             Read more about it in the
4226             [FastAPI docs for Path Operation Configuration]
4227             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
4228             """
4229         ),
4230     ] = None,
4231     response_description: Annotated[
4232         str,
4233         Doc(
4234             """
4235             The description for the default response.
4236
4237             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4238             """
4239         ),
4240     ] = "Successful Response",
4241     responses: Annotated[
4242         dict[int | str, dict[str, Any]] | None,
4243         Doc(
4244             """
4245             Additional responses that could be returned by this *path operation*.
4246
4247             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4248             """
4249         ),
4250     ] = None,
4251     deprecated: Annotated[
4252         bool | None,
4253         Doc(
4254             """
4255             Mark this *path operation* as deprecated.

```

```

4256         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4257         """
4258     ),
4259     ] = None,
4260     operation_id: Annotated[
4261         str | None,
4262         Doc(
4263             """
4264             Custom operation ID to be used by this *path operation*.
4265
4266             By default, it is generated automatically.
4267
4268             If you provide a custom operation ID, you need to make sure it is
4269             unique for the whole API.
4270
4271             You can customize the
4272             operation ID generation with the parameter
4273             `generate_unique_id_function` in the `FastAPI` class.
4274
4275             Read more about it in the
4276             [FastAPI docs about how to Generate Clients]
4277             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4278             """
4279         ),
4280     ] = None,
4281     response_model_include: Annotated[
4282         IncEx | None,
4283         Doc(
4284             """
4285             Configuration passed to Pydantic to include only certain fields in the
4286             response data.
4287
4288             Read more about it in the
4289             [FastAPI docs for Response Model - Return Type]
4290             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4291             """
4292         ),
4293     ] = None,
4294     response_model_exclude: Annotated[
4295         IncEx | None,
4296         Doc(
4297             """
4298             Configuration passed to Pydantic to exclude certain fields in the
4299             response data.
4300
4301             Read more about it in the
4302             [FastAPI docs for Response Model - Return Type]
4303             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4304             """
4305         ),
4306     ] = None,
4307     response_model_by_alias: Annotated[
4308         bool,
4309         Doc(
4310             """
4311             Configuration passed to Pydantic to define if the response model
4312             should be serialized by alias when an alias is used.
4313
4314             Read more about it in the
4315             [FastAPI docs for Response Model - Return Type]
4316             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4317             """
4318         ),
4319     ] = True,
4320     response_model_exclude_unset: Annotated[
4321         bool,
4322         Doc(
4323             """
4324             Configuration passed to Pydantic to define if the response data
4325             should have all the fields, including the ones that were not set and
4326             have their default values. This is different from
4327             `response_model_exclude_defaults` in that if the fields are set,
4328

```

```

4332         they will be included in the response, even if the value is the same
4333         as the default.
4334
4335         When `True`, default values are omitted from the response.
4336
4337         Read more about it in the
4338         [FastAPI docs for Response Model - Return Type]
4339         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4340         parameter).
4341         """
4342     ),
4343     ] = False,
4344     response_model_exclude_defaults: Annotated[
4345         bool,
4346         Doc(
4347             """
4348             Configuration passed to Pydantic to define if the response data
4349             should have all the fields, including the ones that have the same value
4350             as the default. This is different from `response_model_exclude_unset`
4351             in that if the fields are set but contain the same default values,
4352             they will be excluded from the response.
4353
4354             When `True`, default values are omitted from the response.
4355
4356             Read more about it in the
4357             [FastAPI docs for Response Model - Return Type]
4358             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4359             parameter).
4360             """
4361         ),
4362     ] = False,
4363     response_model_exclude_none: Annotated[
4364         bool,
4365         Doc(
4366             """
4367             Configuration passed to Pydantic to define if the response data should
4368             exclude fields set to `None`.
4369
4370             This is much simpler (less smart) than `response_model_exclude_unset`
4371             and `response_model_exclude_defaults`. You probably want to use one of
4372             those two instead of this one, as those allow returning `None` values
4373             when it makes sense.
4374
4375             Read more about it in the
4376             [FastAPI docs for Response Model - Return Type]
4377             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
4378             """
4379         ),
4380     ] = False,
4381     include_in_schema: Annotated[
4382         bool,
4383         Doc(
4384             """
4385             Include this *path operation* in the generated OpenAPI schema.
4386
4387             This affects the generated OpenAPI (e.g. visible at `/docs`).
4388
4389             Read more about it in the
4390             [FastAPI docs for Query Parameters and String Validations]
4391             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
4392             openapi).
4393             """
4394         ),
4395     ] = True,
4396     response_class: Annotated[
4397         type[Response],
4398         Doc(
4399             """
4400             Response class to be used for this *path operation*.
4401
4402             This will not be used if you return a response directly.
4403
4404             Read more about it in the
4405             [FastAPI docs for Custom Response - HTML, Stream, File, others]
4406             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
4407             """

```

```

4408     ),
4409     ] = Default(JSONResponse),
4410     name: Annotated[
4411         str | None,
4412         Doc(
4413             """
4414             Name for this *path operation*. Only used internally.
4415             """
4416         ),
4417     ] = None,
4418     callbacks: Annotated[
4419         list[BaseRoute] | None,
4420         Doc(
4421             """
4422             List of *path operations* that will be used as OpenAPI callbacks.
4423
4424             This is only for OpenAPI documentation, the callbacks won't be used
4425             directly.
4426
4427             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4428
4429             Read more about it in the
4430             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
4431 callbacks/).
4432             """
4433         ),
4434     ] = None,
4435     openapi_extra: Annotated[
4436         dict[str, Any] | None,
4437         Doc(
4438             """
4439             Extra metadata to be included in the OpenAPI schema for this *path
4440             operation*.
4441
4442             Read more about it in the
4443             [FastAPI docs for Path Operation Advanced Configuration]
4444             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
4445             path-operation-schema).
4446             """
4447         ),
4448     ] = None,
4449     generate_unique_id_function: Annotated[
4450         Callable[[APIRoute], str],
4451         Doc(
4452             """
4453             Customize the function used to generate unique IDs for the *path
4454             operations* shown in the generated OpenAPI.
4455
4456             This is particularly useful when automatically generating clients or
4457             SDKs for your API.
4458
4459             Read more about it in the
4460             [FastAPI docs about how to Generate Clients]
4461             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4462             """
4463         ),
4464     ] = Default(generate_unique_id),
4465 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4466     """
4467     Add a *path operation* using an HTTP PATCH operation.
4468
4469     ## Example
4470
4471     ```python
4472     from fastapi import APIRouter, FastAPI
4473     from pydantic import BaseModel
4474
4475     class Item(BaseModel):
4476         name: str
4477         description: str | None = None
4478
4479     app = FastAPI()
4480     router = APIRouter()
4481
4482     @router.patch("/items/")
4483     def update_item(item: Item):

```

```

4484         return {"message": "Item updated in place"}
4485
4486     app.include_router(router)
4487     """
4488
4489     return self.api_route(
4490         path=path,
4491         response_model=response_model,
4492         status_code=status_code,
4493         tags=tags,
4494         dependencies=dependencies,
4495         summary=summary,
4496         description=description,
4497         response_description=response_description,
4498         responses=responses,
4499         deprecated=deprecated,
4500         methods=["PATCH"],
4501         operation_id=operation_id,
4502         response_model_include=response_model_include,
4503         response_model_exclude=response_model_exclude,
4504         response_model_by_alias=response_model_by_alias,
4505         response_model_exclude_unset=response_model_exclude_unset,
4506         response_model_exclude_defaults=response_model_exclude_defaults,
4507         response_model_exclude_none=response_model_exclude_none,
4508         include_in_schema=include_in_schema,
4509         response_class=response_class,
4510         name=name,
4511         callbacks=callbacks,
4512         openapi_extra=openapi_extra,
4513         generate_unique_id_function=generate_unique_id_function,
4514     )

```

meth trace

```

trace(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP TRACE operation.

EXAMPLE

```

from fastapi import APIRouter, FastAPI
from pydantic import BaseModel

class Item(BaseModel):

```



```
name: str
description: str | None = None

app = FastAPI()
router = APIRouter()

@router.trace("/items/{item_id}")
def trace_item(item_id: str):
    return None

app.include_router(router)
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: <code>Any</code> DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration.

PARAMETER	DESCRIPTION		
		TYPE:	DEFAULT:
		<code>list[str Enum] None</code>	<code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>		
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>		
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str None</code> DEFAULT: <code>None</code>		
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>		
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>		
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool None</code> DEFAULT: <code>None</code>		

PARAMETER	DESCRIPTION
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients.

PARAMETER	DESCRIPTION
	TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/routing.py

```

4490 def trace(
4491     self,
4492     path: Annotated[
4493         str,
4494         Doc(
4495             """
4496             The URL path to be used for this *path* operation*.
4497
4498             For example, in `http://example.com/items`, the path is `/items`.
4499             """
4500         ),
4501     ],
4502     *,
4503     response_model: Annotated[
4504         Any,
4505         Doc(
4506             """
4507             The type to use for the response.
4508
4509             It could be any valid Pydantic *field* type. So, it doesn't have to
4510             be a Pydantic model, it could be other things, like a `list`, `dict`,
4511             etc.
4512
4513             It will be used for:
4514
4515             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
4516               show it as the response (JSON Schema).
4517             * Serialization: you could return an arbitrary object and the
4518               `response_model` would be used to serialize that object into the
4519               corresponding JSON.
4520             * Filtering: the JSON sent to the client will only contain the data
4521               (fields) defined in the `response_model`. If you returned an object
4522               that contains an attribute `password` but the `response_model` does
4523               not include that field, the JSON sent to the client would not have
4524               that `password`.
4525             * Validation: whatever you return will be serialized with the
4526               `response_model`, converting any data as necessary to generate the
4527               corresponding JSON. But if the data in the object returned is not
4528               valid, that would mean a violation of the contract with the client,
4529               so it's an error from the API developer. So, FastAPI will raise an
4530               error and return a 500 error code (Internal Server Error).
4531
4532             Read more about it in the
4533             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
4534             model/).
4535             """
4536         ),
4537     ] = Default(None),
4538     status_code: Annotated[
4539         int | None,
4540         Doc(
4541             """
4542             The default status code to be used for the response.
4543
4544             You could override the status code by returning a response directly.
4545
4546             Read more about it in the
4547             [FastAPI docs for Response Status Code]
4548             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
4549             """
4550         ),
4551     ] = None,
4552     tags: Annotated[
4553         list[str | Enum] | None,
4554         Doc(
4555             """
4556             A list of tags to be applied to the *path* operation*.
4557
4558             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4559
4560             Read more about it in the
4561

```

```

4562         [FastAPI docs for Path Operation Configuration]
4563         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
4564         """
4565     ),
4566     ] = None,
4567     dependencies: Annotated[
4568         Sequence[params.Depends] | None,
4569         Doc(
4570             """
4571             A list of dependencies (using `Depends()`) to be applied to the
4572             *path operation*.
4573
4574             Read more about it in the
4575             [FastAPI docs for Dependencies in path operation decorators]
4576             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
4577             """
4578         ),
4579     ] = None,
4580     summary: Annotated[
4581         str | None,
4582         Doc(
4583             """
4584             A summary for the *path operation*.
4585
4586             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4587
4588             Read more about it in the
4589             [FastAPI docs for Path Operation Configuration]
4590             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
4591             """
4592         ),
4593     ] = None,
4594     description: Annotated[
4595         str | None,
4596         Doc(
4597             """
4598             A description for the *path operation*.
4599
4600             If not provided, it will be extracted automatically from the docstring
4601             of the *path operation function*.
4602
4603             It can contain Markdown.
4604
4605             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4606
4607             Read more about it in the
4608             [FastAPI docs for Path Operation Configuration]
4609             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
4610             """
4611         ),
4612     ] = None,
4613     response_description: Annotated[
4614         str,
4615         Doc(
4616             """
4617             The description for the default response.
4618
4619             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4620             """
4621         ),
4622     ] = "Successful Response",
4623     responses: Annotated[
4624         dict[int | str, dict[str, Any]] | None,
4625         Doc(
4626             """
4627             Additional responses that could be returned by this *path operation*.
4628
4629             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4630             """
4631         ),
4632     ] = None,
4633     deprecated: Annotated[
4634         bool | None,
4635         Doc(
4636             """
4637             Mark this *path operation* as deprecated.

```

```

4638         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4639         """
4640     ),
4641     ] = None,
4642     operation_id: Annotated[
4643         str | None,
4644         Doc(
4645             """
4646             Custom operation ID to be used by this *path operation*.
4647
4648             By default, it is generated automatically.
4649
4650             If you provide a custom operation ID, you need to make sure it is
4651             unique for the whole API.
4652
4653             You can customize the
4654             operation ID generation with the parameter
4655             `generate_unique_id_function` in the `FastAPI` class.
4656
4657             Read more about it in the
4658             [FastAPI docs about how to Generate Clients]
4659             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4660             """
4661         ),
4662     ] = None,
4663     response_model_include: Annotated[
4664         IncEx | None,
4665         Doc(
4666             """
4667             Configuration passed to Pydantic to include only certain fields in the
4668             response data.
4669
4670             Read more about it in the
4671             [FastAPI docs for Response Model - Return Type]
4672             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4673             """
4674         ),
4675     ],
4676     response_model_exclude: Annotated[
4677         IncEx | None,
4678         Doc(
4679             """
4680             Configuration passed to Pydantic to exclude certain fields in the
4681             response data.
4682
4683             Read more about it in the
4684             [FastAPI docs for Response Model - Return Type]
4685             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4686             """
4687         ),
4688     ],
4689     response_model_by_alias: Annotated[
4690         bool,
4691         Doc(
4692             """
4693             Configuration passed to Pydantic to define if the response model
4694             should be serialized by alias when an alias is used.
4695
4696             Read more about it in the
4697             [FastAPI docs for Response Model - Return Type]
4698             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4699             """
4700         ),
4701     ],
4702     response_model_exclude_unset: Annotated[
4703         bool,
4704         Doc(
4705             """
4706             Configuration passed to Pydantic to define if the response data
4707             should have all the fields, including the ones that were not set and
4708             have their default values. This is different from
4709             `response_model_exclude_defaults` in that if the fields are set,
4710

```



```

4714         they will be included in the response, even if the value is the same
4715         as the default.
4716
4717         When `True`, default values are omitted from the response.
4718
4719         Read more about it in the
4720         [FastAPI docs for Response Model - Return Type]
4721         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4722         parameter).
4723         """
4724     ),
4725     ] = False,
4726     response_model_exclude_defaults: Annotated[
4727         bool,
4728         Doc(
4729             """
4730             Configuration passed to Pydantic to define if the response data
4731             should have all the fields, including the ones that have the same value
4732             as the default. This is different from `response_model_exclude_unset`
4733             in that if the fields are set but contain the same default values,
4734             they will be excluded from the response.
4735
4736             When `True`, default values are omitted from the response.
4737
4738             Read more about it in the
4739             [FastAPI docs for Response Model - Return Type]
4740             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4741             parameter).
4742             """
4743         ),
4744     ] = False,
4745     response_model_exclude_none: Annotated[
4746         bool,
4747         Doc(
4748             """
4749             Configuration passed to Pydantic to define if the response data should
4750             exclude fields set to `None`.
4751
4752             This is much simpler (less smart) than `response_model_exclude_unset`
4753             and `response_model_exclude_defaults`. You probably want to use one of
4754             those two instead of this one, as those allow returning `None` values
4755             when it makes sense.
4756
4757             Read more about it in the
4758             [FastAPI docs for Response Model - Return Type]
4759             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
4760             """
4761         ),
4762     ] = False,
4763     include_in_schema: Annotated[
4764         bool,
4765         Doc(
4766             """
4767             Include this *path operation* in the generated OpenAPI schema.
4768
4769             This affects the generated OpenAPI (e.g. visible at `/docs`).
4770
4771             Read more about it in the
4772             [FastAPI docs for Query Parameters and String Validations]
4773             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
4774             openapi).
4775             """
4776         ),
4777     ] = True,
4778     response_class: Annotated[
4779         type[Response],
4780         Doc(
4781             """
4782             Response class to be used for this *path operation*.
4783
4784             This will not be used if you return a response directly.
4785
4786             Read more about it in the
4787             [FastAPI docs for Custom Response - HTML, Stream, File, others]
4788             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
4789             """

```

```

4790     ),
4791     ] = Default(JSONResponse),
4792     name: Annotated[
4793         str | None,
4794         Doc(
4795             """
4796             Name for this *path operation*. Only used internally.
4797             """
4798         ),
4799     ] = None,
4800     callbacks: Annotated[
4801         list[BaseRoute] | None,
4802         Doc(
4803             """
4804             List of *path operations* that will be used as OpenAPI callbacks.
4805
4806             This is only for OpenAPI documentation, the callbacks won't be used
4807             directly.
4808
4809             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4810
4811             Read more about it in the
4812             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
4813             callbacks/).
4814             """
4815         ),
4816     ] = None,
4817     openapi_extra: Annotated[
4818         dict[str, Any] | None,
4819         Doc(
4820             """
4821             Extra metadata to be included in the OpenAPI schema for this *path
4822             operation*.
4823
4824             Read more about it in the
4825             [FastAPI docs for Path Operation Advanced Configuration]
4826             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
4827             path-operation-schema).
4828             """
4829         ),
4830     ] = None,
4831     generate_unique_id_function: Annotated[
4832         Callable[[APIRoute], str],
4833         Doc(
4834             """
4835             Customize the function used to generate unique IDs for the *path
4836             operations* shown in the generated OpenAPI.
4837
4838             This is particularly useful when automatically generating clients or
4839             SDKs for your API.
4840
4841             Read more about it in the
4842             [FastAPI docs about how to Generate Clients]
4843             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4844             """
4845         ),
4846     ] = Default(generate_unique_id),
4847 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4848     """
4849     Add a *path operation* using an HTTP TRACE operation.
4850
4851     ## Example
4852
4853     ```python
4854     from fastapi import APIRouter, FastAPI
4855     from pydantic import BaseModel
4856
4857     class Item(BaseModel):
4858         name: str
4859         description: str | None = None
4860
4861     app = FastAPI()
4862     router = APIRouter()
4863
4864     @router.trace("/items/{item_id}")
4865     def trace_item(item_id: str):

```

```
4866         return None
4867
4868     app.include_router(router)
4869     """
4870     return self.api_route(
        path=path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        methods=["TRACE"],
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )
```

meth on_event

```
on_event(event_type)
```

Add an event handler for the router.

`on_event` is deprecated, use `lifespan` event handlers instead.

Read more about it in the [FastAPI docs for Lifespan Events](#).

PARAMETER	DESCRIPTION
<code>event_type</code>	The type of event. <code>startup</code> or <code>shutdown</code> . TYPE: <code>str</code>

< > Source code in `fastapi/routing.py`

```
4924 @deprecated(
4925     """
4926     on_event is deprecated, use lifespan event handlers instead.
4927
4928     Read more about it in the
4929     [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/events/).
4930     """
4931 )
4932 def on_event(
4933     self,
4934     event_type: Annotated[
4935         str,
4936         Doc(
4937             """
4938             The type of event. `startup` or `shutdown`.
4939             """
4940         ),
4941     ],
4942 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4943     """
4944     Add an event handler for the router.
4945
4946     `on_event` is deprecated, use `lifespan` event handlers instead.
4947
4948     Read more about it in the
4949     [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/events/#alternative-
4950     events-deprecated).
4951     """
4952
4953     def decorator(func: DecoratedCallable) -> DecoratedCallable:
4954         self.add_event_handler(event_type, func)
4955         return func
4956
4957     return decorator
```